

Beyond Minimum Distance: The Optimal Leading Coefficient in the High-SNR Error-Probability Expansion for AWGN Spherical Codes

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Abstract

Packing-optimal (M, n) spherical codes attain the largest achievable minimum distance and therefore achieve the optimal exponential decay rate of the error probability at high signal-to-noise ratio (SNR). Among packing-optimal codebooks held fixed as the SNR grows, the smallest leading coefficient is $K_{M,n}^{\text{fix}}$, equal to the smallest number of ordered closest pairs of codewords among packing-optimal codebooks. We show that SNR-wise codebook optimization can achieve a smaller leading coefficient, and thereby a smaller error probability, than any packing-optimal codebook held fixed. Let $P_e^*(M, n; \gamma)$ be the SNR-wise minimum exact maximum-likelihood (ML) error probability and let $P_e^{\text{fix}}(M, n; \gamma)$ be the minimum exact error over all packing-optimal codebooks. We prove that $P_e^* = B_\gamma^*(K_{M,n}^* + o(1))$ and $P_e^{\text{fix}} = B_\gamma^*(K_{M,n}^{\text{fix}} + o(1))$, and that $K_{M,n}^* \leq K_{M,n}^{\text{fix}}$, where B_γ^* is the common Gaussian-tail factor. Hence, whenever $K_{M,n}^* < K_{M,n}^{\text{fix}}$, the SNR-wise minimum exact error is strictly smaller than the best packing-optimal benchmark at all sufficiently high SNRs, i.e., $P_e^* < P_e^{\text{fix}}$. We also show that the minimum of the Gaussian soft-packing energy converges to $K_{M,n}^*$.

The orthoplex-bound construction reveals how $K_{M,n}^* < K_{M,n}^{\text{fix}}$ can occur. The construction splits the pairs that are closest in the limiting packing into two classes. At finite SNR, the pairs in the first class are deliberately made slightly closer than the optimal packing distance. This change is too small to affect either the optimal asymptotic error exponent or the selected pairs' unit contributions to the constructed family's leading coefficient. However, this creates enough geometric freedom to move the other pairs—those in the second class—farther apart on a larger but still vanishing scale, so that their contributions to the constructed family's leading coefficient vanish. By contrast, every closest pair

of a packing-optimal codebook held fixed contributes one to that codebook's fixed-codebook leading coefficient.

We apply the developed theoretical framework to the simplex range, $2 \leq M \leq n+1$, and the orthoplex-bound range, $M = n + k$, $2 \leq k \leq n$. We obtain $K_{M,n}^* = K_{M,n}^{\text{fix}} = M(M-1)$ for $2 \leq M \leq n+1$. For $M = n + k$, $2 \leq k \leq n$, we prove $K_{n+k,n}^{\text{fix}} = 4n(k-1)$ and $K_{n+k,n}^* \leq 4k(k-1)$. For $n \geq 3$ and $M = n + 2$, $K_{n+2,n}^* = 8 < 4n = K_{n+2,n}^{\text{fix}}$, so the optimized error is asymptotically smaller by the factor $n/2$ than the best packing-optimal error. At the terminal endpoint $k = n$, $K_{2n,n}^* = K_{2n,n}^{\text{fix}} = 4n(n-1)$. For $3 \leq k \leq n-1$, we conjecture that $K_{n+k,n}^* = 4k(k-1)$.

Index Terms

AWGN channels, spherical codes, maximum-likelihood decoding, message error probability, high-SNR asymptotics, spherical packing, codebook optimization, leading coefficients.

I. INTRODUCTION

At high average signal-to-noise ratio (SNR) per real channel use, finite spherical-code design raises two nested questions. First, how rapidly can the minimum achievable average message error probability under maximum-likelihood (ML) decoding decay? Second, after removing that decay scale, how small can the remaining leading coefficient be? Optimal spherical packing, through the largest achievable minimum distance, answers the first question. It need not answer the second. Codebook families can have the same packing-determined exponential decay yet differ by a nonvanishing multiplicative factor in error probability.

For a packing-optimal codebook held fixed as the SNR grows, classical high-SNR theory shows that its number of ordered closest pairs of codewords determines the leading coefficient in its high-SNR error-probability expansion [1], [2]. Dividing this count by M gives the codebook's average kissing number. Hence, among packing-optimal codebooks that are each held fixed as the SNR grows, minimizing the leading coefficient is equivalent to choosing a codebook with the smallest ordered closest-pair count, or, equivalently, the smallest average kissing number. The optimization studied here is different: the codeword locations may be reoptimized at every SNR. Because pairwise Gaussian error probabilities depend exponentially on the SNR and the pairwise squared distances, vanishing changes in codeword distances can still change the error probability by a nonvanishing factor. This allows an SNR-wise optimized codebook family to approach the set of optimal packings while having a smaller leading coefficient than every packing-optimal codebook held fixed. Its error probability is then smaller than the best fixed packing-optimal

benchmark at all sufficiently high SNRs. Consequently, optimizing the codebook at each SNR and then taking the high-SNR limit can yield a smaller leading coefficient than first taking the high-SNR limit for each fixed codebook and then optimizing over the packing-optimal class.

The explicit SNR-dependent construction developed in this paper for the orthoplex-bound range $M = n + k$, $2 \leq k < n$, makes this phenomenon concrete. The construction splits the codeword pairs that are closest in the limiting packing into two classes. At finite SNR, the pairs in the first class are deliberately made slightly closer than the optimal packing distance. This deterioration is too small to change either the optimal asymptotic error exponent or the selected pairs' unit contributions to the constructed family's leading coefficient. However, it creates enough geometric freedom to move the other pairs—those in the second class—farther apart on a larger but still vanishing scale, so that their contributions to the constructed family's leading coefficient vanish. The constructed family retains the packing-optimal exponent and approaches an optimal packing as the SNR grows, yet lies outside the set of optimal packings at every sufficiently large finite SNR. Its leading coefficient is smaller than that of every packing-optimal codebook held fixed as the SNR grows. This improvement cannot be obtained by holding any one codebook fixed. The smaller coefficient is obtained only because the codebook continues to change with SNR.

Let $P_e^*(M, n; \gamma)$ denote the exact average ML message error probability minimized over all (M, n) spherical codebooks at SNR γ . For every fixed finite pair (M, n) considered here, we prove the general expansion

$$P_e^*(M, n; \gamma) = B_\gamma^*(K_{M,n}^* + o(1)),$$

where B_γ^* contains the packing-determined exponential decay and Gaussian prefactor, and $K_{M,n}^*$ is a well-defined optimal leading coefficient.

For comparison, let $P_e^{\text{fix}}(M, n; \gamma)$ denote the exact average ML message error probability minimized over all spherical codebooks attaining the largest possible minimum distance. Then

$$P_e^{\text{fix}}(M, n; \gamma) = B_\gamma^*(K_{M,n}^{\text{fix}} + o(1)),$$

where $K_{M,n}^{\text{fix}}$ is the smallest number of ordered closest-pair indices among packing-optimal codebooks and $K_{M,n}^{\text{fix}}/M$ is the smallest average kissing number. We show that $K_{M,n}^* \leq K_{M,n}^{\text{fix}}$. If $K_{M,n}^* < K_{M,n}^{\text{fix}}$, then $P_e^*(M, n; \gamma) < P_e^{\text{fix}}(M, n; \gamma)$ for all sufficiently large γ , and

$$\frac{P_e^*(M, n; \gamma)}{P_e^{\text{fix}}(M, n; \gamma)} \longrightarrow \frac{K_{M,n}^*}{K_{M,n}^{\text{fix}}} < 1.$$

Thus the improvement is a persistent multiplicative reduction in the minimum exact error probability.

We call an SNR-dependent family *packing-tail competitive* if its normalized error remains bounded on the packing-tail scale. We show that every such family approaches the set of optimal packings. If such a family has a leading coefficient, then, along any convergent codebook subsequence, only pairs that are closest in the packing-optimal limiting codebook can contribute to that coefficient. The contribution of each such pair depends on the rate at which its correlation approaches the optimal packing value and may be zero, fractional, one, or greater than one. For a family attaining the optimal leading coefficient, these pairwise contributions sum to $K_{M,n}^*$. Thus, for a family attaining the optimal leading coefficient, the packing-optimal limiting codebook identifies which pairs can contribute to $K_{M,n}^*$, whereas the SNR-dependent codebook optimization determines how much each such pair contributes to $K_{M,n}^*$.

To analyze this geometry, we introduce a Gaussian soft-packing energy that continuously accounts for all pairwise confusions. We prove that it uniformly represents the normalized exact ML error for the SNR-dependent families relevant to optimized high-SNR design, and that its minimum converges to $K_{M,n}^*$. Consequently, every family whose Gaussian soft-packing energy is within an additive $o(1)$ of its global minimum is asymptotically ML-optimal.

A. Contributions

The contributions are as follows. The first three contributions apply to every fixed finite (M, n) (with $M \geq 2$) for which the best packing has nonzero minimum distance. The last two apply the theory throughout the simplex and orthoplex-bound ranges.

- 1) *Optimal leading-coefficient characterization and fixed-codebook benchmark.* We introduce a Gaussian soft-packing energy and prove that it uniformly represents the normalized exact ML error on the bounded Gaussian soft-packing energy classes relevant to packing-tail-competitive designs. Its minimum converges to a well-defined optimal leading coefficient $K_{M,n}^*$, yielding the high-SNR expansion of the minimum average error probability, $P_e^*(M, n; \gamma) = B_\gamma^*(K_{M,n}^* + o(1))$. We also prove $P_e^{\text{fix}}(M, n; \gamma) = B_\gamma^*(K_{M,n}^{\text{fix}} + o(1))$ for the best packing-optimal benchmark. We show that $K_{M,n}^* \leq K_{M,n}^{\text{fix}}$. Hence, when $K_{M,n}^* < K_{M,n}^{\text{fix}}$, $P_e^*(M, n; \gamma) < P_e^{\text{fix}}(M, n; \gamma)$ for all sufficiently large γ , and their ratio tends to $K_{M,n}^*/K_{M,n}^{\text{fix}}$ as the SNR grows.

- 2) *Localization and SNR-dependent closest-pair contributions of packing-tail-competitive families.* We prove that every family satisfying $P_e(\mathcal{C}_\gamma; \gamma)/B_\gamma^* = O(1)$ approaches the packing-optimal set. Along convergent subsequences, only pairs that are closest in the packing-optimal limiting codebook can have nonvanishing normalized contributions, and these contributions are determined by their scaled correlation offsets. Whenever the family's leading coefficient exists, these contributions sum to that coefficient; for a family attaining the optimal leading coefficient, they sum to $K_{M,n}^*$. This explains why neither an optimal packing nor its ordinary closest-pair count alone determines the optimal leading coefficient $K_{M,n}^*$. Minimizing over all packing-optimal limiting codebooks and all jointly feasible scaled closest-pair offsets yields an exact variational characterization of $K_{M,n}^*$.
- 3) *Attainment and matching-bound certification.* We prove that codebook families whose Gaussian soft-packing energy is within an additive $o(1)$ of its global minimum are asymptotically ML-optimal. We then give a matching-bounds criterion that certifies both a proposed optimal leading coefficient and a full SNR-indexed construction attaining the optimal leading coefficient.
- 4) *Simplex code.* For every $2 \leq M \leq n + 1$, we prove that an embedded regular $(M - 1)$ -simplex minimizes the soft-packing objective exactly at every positive SNR and that $K_{M,n}^* = K_{M,n}^{\text{fix}} = M(M - 1)$. The constant simplex family is therefore asymptotically ML-optimal under SNR-wise reoptimization.
- 5) *Orthoplex-bound range.* For the orthoplex-bound range $M = n + k$, $2 \leq k \leq n$, we determine $K_{n+k,n}^{\text{fix}} = 4n(k - 1)$. For every $2 \leq k < n$, an explicit SNR-dependent family has leading coefficient $4k(k - 1)$, which proves $K_{n+k,n}^* \leq 4k(k - 1)$. Hence, the optimal leading coefficient, $K_{n+k,n}^*$, is strictly smaller than the fixed benchmark, $K_{n+k,n}^{\text{fix}}$, and

$$\lim_{\gamma \rightarrow \infty} \frac{P_e^*(n + k, n; \gamma)}{P_e^{\text{fix}}(n + k, n; \gamma)} = \frac{K_{n+k,n}^*}{K_{n+k,n}^{\text{fix}}} \leq \frac{k}{n} < 1.$$

At the left endpoint $k = 2$, a matching converse proves $K_{n+2,n}^* = 8$, thereby showing that $P_e^*(n + 2, n; \gamma)$ is asymptotically smaller by the factor $n/2$ than $P_e^{\text{fix}}(n + 2, n; \gamma)$. At the terminal endpoint $k = n$, universal optimality of the full cross-polytope proves $K_{2n,n}^* = K_{2n,n}^{\text{fix}} = 4n(n - 1)$. For $3 \leq k \leq n - 1$, we conjecture that $K_{n+k,n}^* = 4k(k - 1)$.

B. Relation to Existing Theory

For a fixed finite codebook over the AWGN channel, the pairwise error is a Gaussian tail, and the closest-pair expansion is asymptotically equivalent to the exact ML message error probability [1], [2]. Because these results hold the codebook fixed, they do not justify minimizing the resulting formula over a different codebook at every SNR. At the exponential scale, equal-energy codebook design is the classical spherical-packing problem [3]–[6], for which linear- and semidefinite-programming methods provide powerful general bounds [7]–[10].

The simplex application uses the simplex bound [4] and the universal optimality of embedded regular simplices [11]. The orthoplex-bound application uses the orthoplex bound [4] and its equality-case classification [12, Thm. 3]; an equivalent characterization is given in [13, Thm. 2.2]. It also uses the universal optimality of the full cross-polytope [11]. The recent result in [14, Cors. 2.5–2.6] proves the stronger finite-SNR coding statement that, for M equiprobable equal-energy signals over the real AWGN channel, an embedded regular $(M - 1)$ -simplex maximizes the average probability of correct ML decoding at every positive SNR whenever $n \geq M - 1$. That result also characterizes the equality case: the embedded regular simplex is the unique maximizer up to an orthogonal transformation and relabeling.

A closely related large-parameter phenomenon occurs for Riesz energies. In [15], the authors proved that cluster points of fixed-cardinality Riesz-energy minimizers as the exponent tends to infinity are best packings. For five points on $\mathbb{S}^2$, they identified, up to isometry, the unique limiting square-pyramid configuration and proved that the normalized minimum Riesz energy converges to 8; their upper bound uses an exponent-dependent square-pyramid deformation. This is a close antecedent to the $(M, n) = (5, 3)$ specialization of the SNR-dependent construction in our work. Their Riesz objective, however, neither gives the Gaussian construction for general $(n + k, n)$ nor transfers its asymptotics to exact AWGN ML error.

In a different positive-temperature cross-entropy problem motivated by neural collapse, [16] show that, among packing-optimal codes in the orthoplex regime, sufficiently low temperature selects a “low-entropy” geometry consisting of one regular simplex and an orthogonal cross-polytope. For $M = n + k$, this is the same simplex-plus-antipodal-pairs geometry that attains the best fixed-codebook leading coefficient here. Their feasible class and objective are different, and their result does not address SNR-dependent paths outside the packing-optimal set.

The preceding works provide geometric, coding, and energy results that serve as technical inputs

or conceptual antecedents for particular simplex and orthoplex-bound applications considered here. The theory developed in this paper is broader: it is not restricted to these special geometries and characterizes the optimal high-SNR leading coefficient when a finite spherical codebook may be reoptimized at each SNR. Its new ingredients are the uniform transfer from the soft-packing energy to exact ML error, the existence and characterization of the optimal leading coefficient, and the localization and closest-pair contribution results for competitive SNR-dependent families. The simplex and orthoplex-bound results are concrete applications of this general framework.

C. Organization

Section II formulates the exact design problem and its fixed-codebook benchmark. Section III develops the soft-packing characterization of the optimal leading coefficient and establishes attainment by near-minimizers of the Gaussian soft-packing energy. Section IV establishes localization, the SNR-dependent closest-pair structure of competitive families, and the global variational characterization by jointly feasible scaled offsets. Section V develops the matching-bounds criterion used in the applications. Sections VI and VII apply this theory to the simplex and orthoplex-bound ranges, respectively. Section VIII concludes the paper, and the appendices provide the technical proofs.

II. SIGNAL MODEL AND HIGH-SNR CODEBOOK DESIGN PROBLEM

This section introduces the exact AWGN error probability, the spherical-packing quantities that determine its optimal exponential decay, and the classical high-SNR expansion for a fixed codebook. It then defines the exact-error benchmark obtained by restricting the design to packing-optimal codebooks and formulates the unrestricted optimization performed separately at each SNR.

A. Signal Model and Exact Error Probability

Fix integers $M \geq 2$ and $n \geq 1$, write $[M] = \{1, \dots, M\}$, and let $\mathbb{S}^{n-1} = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\}$. A labeled spherical codebook is

$$\mathcal{C} = (\mathbf{c}_1, \dots, \mathbf{c}_M) \in \mathcal{M}, \quad \mathcal{M} = (\mathbb{S}^{n-1})^M,$$

where every codeword has unit norm. The product space $\mathcal{M}$ is compact.

Let W be uniform on $[M]$. The normalized real AWGN channel is

$$\mathbf{Y} = \sqrt{n\gamma} \mathbf{c}_W + \mathbf{Z}, \quad \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n). \quad (1)$$

The noise variance is one per real channel use, and the transmitted vector has energy $n\gamma$; hence γ is the average SNR per real channel use. Unless stated otherwise, every high-SNR limit keeps (M, n) fixed and lets $\gamma \rightarrow \infty$.

For an observation $\mathbf{y} \in \mathbb{R}^n$, the maximum-likelihood decoder is

$$\widehat{W}(\mathbf{y}) \in \arg \max_{1 \leq m \leq M} \mathbf{y}^\top \mathbf{c}_m. \quad (2)$$

Because all codewords have the same norm, maximizing correlation is equivalent to minimizing Euclidean distance. We use an arbitrary fixed measurable rule to break ties. With equiprobable messages, the average error is independent of that choice, and ties have probability zero when no two codewords coincide.

The exact average message error probability is

$$P_e(\mathcal{C}; \gamma) = \frac{1}{M} \sum_{m=1}^M \mathbb{P}\{\widehat{W}(\mathbf{Y}) \neq m \mid W = m\}. \quad (3)$$

B. Pairwise Geometry and the Packing Benchmark

Let

$$\mathcal{E}_M = \{(m, \ell) : m, \ell \in [M], m \neq \ell\}$$

be the ordered-pair index set. For $e = (m, \ell) \in \mathcal{E}_M$, define the pairwise correlation

$$\rho_e(\mathcal{C}) = \mathbf{c}_m^\top \mathbf{c}_\ell. \quad (4)$$

Because the codewords have unit norm,

$$\|\mathbf{c}_m - \mathbf{c}_\ell\|^2 = 2(1 - \rho_e(\mathcal{C})). \quad (5)$$

Thus a larger correlation means a smaller Euclidean distance, and minimizing the largest pairwise correlation is equivalent to solving the spherical-code, or spherical-cap-packing, problem [4]–[6].

The largest correlation of a particular codebook and the smallest such largest correlation attainable over all codebooks are, respectively,

$$\rho_{\max}(\mathcal{C}) = \max_{e \in \mathcal{E}_M} \rho_e(\mathcal{C}), \quad \rho_{M,n}^* = \min_{\mathcal{C} \in \mathcal{M}} \rho_{\max}(\mathcal{C}). \quad (6)$$

The minimum exists because $\mathcal{M}$ is compact and $\rho_{\max}$ is continuous. Thus $\rho_{\max}(\mathcal{C})$ is the worst correlation within the particular codebook $\mathcal{C}$: it identifies its closest pair or pairs. By contrast, $\rho_{M,n}^*$ is the best worst correlation achievable among all (M, n) spherical codebooks. Equivalently, $2(1 - \rho_{M,n}^*)$ is the largest achievable minimum squared distance.

The packing-optimal codebook set is

$$\mathcal{P}^* = \{\mathcal{C} \in \mathcal{M} : \rho_{\max}(\mathcal{C}) = \rho_{M,n}^*\}. \quad (7)$$

Hence, $\mathcal{P}^*$ is the collection of all codebooks whose worst correlation equals the best worst correlation $\rho_{M,n}^*$. It contains every solution of the spherical-packing problem, including rotated and relabeled copies and, when they exist, geometrically noncongruent optimal packings.

The analysis in this paper concerns spherical codes for which $\rho_{M,n}^* < 1$. Then every packing-optimal codebook has pairwise-distinct codewords, and its optimal minimum squared distance is $2(1 - \rho_{M,n}^*) > 0$. The value $\rho_{M,n}^*$ determines the best high-SNR error exponent.

An unordered codeword pair $\{m, \ell\}$ is a *closest pair* of $\mathcal{C}$ if it attains the minimum intercodeword distance, or equivalently if $\rho_{(m,\ell)}(\mathcal{C}) = \rho_{\max}(\mathcal{C})$. Collect the corresponding ordered closest-pair indices in

$$\mathcal{I}_{\max}(\mathcal{C}) = \{e \in \mathcal{E}_M : \rho_e(\mathcal{C}) = \rho_{\max}(\mathcal{C})\}. \quad (8)$$

For each unordered closest pair, this set contains both directed indices (m, ℓ) and (ℓ, m) . If $\mathcal{C} \in \mathcal{P}^*$, every index in $\mathcal{I}_{\max}(\mathcal{C})$ has correlation $\rho_{M,n}^*$.

C. Fixed-Codebook High-SNR Asymptotics

Assume a codebook $\mathcal{C}$ is *pairwise distinct*, i.e., $\rho_{\max}(\mathcal{C}) < 1$ holds. For such a codebook and distinct m, ℓ , conditioned on $W = m$, let

$$\mathcal{A}_{m\ell} = \{(\sqrt{n\gamma} \mathbf{c}_m + \mathbf{Z})^\top \mathbf{c}_\ell \geq (\sqrt{n\gamma} \mathbf{c}_m + \mathbf{Z})^\top \mathbf{c}_m\}$$

be the event that competitor ℓ scores at least as highly as the transmitted codeword. For $e = (m, \ell)$, write $\mathcal{A}_e := \mathcal{A}_{m\ell}$. The standard pairwise Gaussian calculation gives

$$\mathbb{P}(\mathcal{A}_{m\ell}) = Q\left(\sqrt{\frac{n\gamma[1 - \rho_{(m,\ell)}(\mathcal{C})]}{2}}\right), \quad (9)$$

where $Q(x) = (2\pi)^{-1/2} \int_x^\infty \exp\{-t^2/2\} dt$ [1]. Consequently,

$$P_e(\mathcal{C}; \gamma) = \frac{1}{M} \sum_{m=1}^M \mathbb{P}\left(\bigcup_{\ell \neq m} \mathcal{A}_{m\ell}\right). \quad (10)$$

For $\rho < 1$, define the single-ordered-pair tail scale

$$b_\gamma(\rho) = \frac{\exp\{-n\gamma(1-\rho)/4\}}{M\sqrt{n\gamma}\sqrt{\pi(1-\rho)}}. \quad (11)$$

The standard fixed-codebook high-SNR result then takes the following form.

Proposition 1 (Fixed-codebook high-SNR expansion). *Let $\mathcal{C}$ be fixed with pairwise-distinct codewords, equivalently $\rho_{\max}(\mathcal{C}) < 1$. Then*

$$\begin{aligned} P_e(\mathcal{C}; \gamma) &= \frac{|\mathcal{I}_{\max}(\mathcal{C})|}{M} Q\left(\sqrt{\frac{n\gamma[1-\rho_{\max}(\mathcal{C})]}{2}}\right) (1 + o_{\mathcal{C}}(1)) \\ &= b_\gamma(\rho_{\max}(\mathcal{C}))(|\mathcal{I}_{\max}(\mathcal{C})| + o_{\mathcal{C}}(1)). \end{aligned} \quad (12)$$

Consequently,

$$\lim_{\gamma \rightarrow \infty} -\gamma^{-1} \log P_e(\mathcal{C}; \gamma) = \frac{n[1-\rho_{\max}(\mathcal{C})]}{4}. \quad (13)$$

Here and throughout, $\log$ denotes the natural logarithm.

The proposition follows by specializing the standard fixed-constellation result [2, Th. 3] to equal-prior spherical codebooks. It is the conventional nearest-neighbor principle stated directly as an expansion of the exact error: after normalization by the tail scale associated with its own worst correlation, a fixed codebook has leading coefficient $|\mathcal{I}_{\max}(\mathcal{C})|$. The remainder $o_{\mathcal{C}}(1)$ is pointwise in $\mathcal{C}$: it tends to zero for each fixed codebook, but its rate may vary with the codebook.

Evaluating (11) at the optimal packing correlation defines the single-ordered-pair packing-tail scale

$$B_\gamma^* := b_\gamma(\rho_{M,n}^*) = \frac{\exp\{-n\gamma(1-\rho_{M,n}^*)/4\}}{M\sqrt{n\gamma}\sqrt{\pi(1-\rho_{M,n}^*)}}. \quad (14)$$

It contains the optimal exponential decay, the Gaussian-tail prefactor, and the $1/M$ message average, but no closest-pair multiplicity. Any fixed codebook outside $\mathcal{P}^*$ has $\rho_{\max}(\mathcal{C}) > \rho_{M,n}^*$, and its exact error divided by B_γ^* therefore grows exponentially. Hence only packing-optimal codebooks can serve as fixed-codebook benchmarks at the optimal exponent, and every fixed $\mathcal{C}_0 \in \mathcal{P}^*$ satisfies

$$\frac{P_e(\mathcal{C}_0; \gamma)}{B_\gamma^*} \longrightarrow |\mathcal{I}_{\max}(\mathcal{C}_0)|. \quad (15)$$

The best fixed-codebook leading coefficient attainable among packing-optimal codebooks is therefore

$$K_{M,n}^{\text{fix}} := \min_{\mathcal{C}_0 \in \mathcal{P}^*} |\mathcal{I}_{\max}(\mathcal{C}_0)|. \quad (16)$$

Equivalently, among all packing-optimal codebooks, one chooses a codebook with the fewest ordered closest-pair indices and uses that same codebook as the SNR increases. Moreover, $K_{M,n}^{\text{fix}}/M$ is the smallest average kissing number among packing-optimal codebooks.

To compare exact error probabilities, define the best packing-optimal benchmark at SNR γ by

$$P_e^{\text{fix}}(M, n; \gamma) := \min_{\mathcal{C}_0 \in \mathcal{P}^*} P_e(\mathcal{C}_0; \gamma). \quad (17)$$

The minimum is attained because $\mathcal{P}^*$ is compact and the exact average ML error is continuous on this set. We show later, cf. Corollary 1, that

$$P_e^{\text{fix}}(M, n; \gamma) = B_\gamma^*(K_{M,n}^{\text{fix}} + o(1)). \quad (18)$$

Accordingly, $K_{M,n}^{\text{fix}}$ and $P_e^{\text{fix}}(M, n; \gamma)$ are the best fixed-codebook leading-coefficient and exact-error benchmarks against which the SNR-wise optimized quantities are compared.

D. Exact High-SNR Codebook Design Problem

For fixed (M, n) and operating SNR γ , the exact SNR-wise optimal average error probability is

$$P_e^*(M, n; \gamma) = \min_{\mathcal{C} \in \mathcal{M}} P_e(\mathcal{C}; \gamma). \quad (19)$$

Thus the minimization ranges over all labeled M -tuples of unit vectors in $\mathbb{R}^n$, and $P_e^*(M, n; \gamma)$ denotes the smallest exact average ML error probability among them. The minimum is attained because $\mathcal{M}$ is compact and the average ML error is continuous in the codeword tuple.

An optimizing codebook may be denoted by

$$\mathcal{C}_\gamma^* \in \arg \min_{\mathcal{C} \in \mathcal{M}} P_e(\mathcal{C}; \gamma).$$

The subscript emphasizes that the optimizing codebook may change with SNR; there is no requirement that $\mathcal{C}_{\gamma_1}^* = \mathcal{C}_{\gamma_2}^*$.

Because $\mathcal{P}^* \subseteq \mathcal{M}$, the two exact design problems in (17) and (19) obey

$$P_e^*(M, n; \gamma) \leq P_e^{\text{fix}}(M, n; \gamma) \quad \text{for every } \gamma. \quad (20)$$

Every fixed $\mathcal{C}_0 \in \mathcal{P}^*$ achieves the best error exponent. Among such fixed codebooks, however, only a minimizer in (16) is guaranteed to have the smallest fixed-codebook leading coefficient. Determining the unrestricted exact finite-SNR minimum $P_e^*(M, n; \gamma)$ and its optimal leading coefficient requires optimization over the full codebook manifold.

The central problem of this paper is therefore to determine whether the limit

$$\lim_{\gamma \rightarrow \infty} \frac{P_e^*(M, n; \gamma)}{B_\gamma^*} \quad (21)$$

exists and is finite and, if so, to characterize it. Once existence is proved, the limit is denoted by $K_{M,n}^*$ and is called the *optimal leading coefficient*. For any SNR-dependent family, if $P_e(\mathcal{C}_\gamma; \gamma)/B_\gamma^*$ converges to a finite limit, we call that limit the *leading coefficient of the family*. For a general SNR-dependent family on which no asymptotic optimality condition is imposed, this finite limit need not exist: the normalized error may be unbounded, or it may remain bounded while approaching different finite values along different SNR subsequences. This possible nonexistence concerns the leading coefficient of such a family and does not arise for the optimal leading coefficient. As shown later, cf. Theorem 2, $P_e^*(M, n; \gamma)/B_\gamma^*$ converges to a finite limit and hence the optimal leading coefficient $K_{M,n}^*$ exists.

An SNR-dependent family $\{\mathcal{C}_\gamma\}_\gamma$ attains the optimal leading coefficient if it satisfies

$$\begin{aligned} \frac{P_e(\mathcal{C}_\gamma; \gamma)}{B_\gamma^*} &\longrightarrow K_{M,n}^*, \\ \frac{P_e(\mathcal{C}_\gamma; \gamma)}{P_e^*(M, n; \gamma)} &\longrightarrow 1. \end{aligned} \quad (22)$$

A second objective is to investigate whether $K_{M,n}^* < K_{M,n}^{\text{fix}}$ can occur and to characterize the associated geometric mechanism. A strict inequality means that SNR-dependent motion of the codewords toward the packing-optimal set improves the optimal leading coefficient even though both designs have the same optimal error exponent. Together with (18), the main optimized expansion proved below then gives

$$\frac{P_e^*(M, n; \gamma)}{P_e^{\text{fix}}(M, n; \gamma)} \longrightarrow \frac{K_{M,n}^*}{K_{M,n}^{\text{fix}}}. \quad (23)$$

Hence $K_{M,n}^* < K_{M,n}^{\text{fix}}$ implies $P_e^*(M, n; \gamma) < P_e^{\text{fix}}(M, n; \gamma)$ for all sufficiently large γ .

E. How $K_{M,n}^* < K_{M,n}^{\text{fix}}$ Can Occur

The best fixed-codebook leading coefficient $K_{M,n}^{\text{fix}}$ and the optimal leading coefficient $K_{M,n}^*$ arise from different high-SNR procedures. The coefficient $K_{M,n}^{\text{fix}}$ is obtained by holding a packing-optimal codebook fixed as the SNR grows, identifying its number of ordered closest-pair indices, and then minimizing this number over $\mathcal{P}^*$. By contrast, $K_{M,n}^*$ is attained asymptotically by an SNR-dependent family $\{\mathcal{C}_\gamma\}_\gamma$. By definition, no codebook in $\mathcal{P}^*$ has fewer than $K_{M,n}^{\text{fix}}$ ordered closest-pair indices. Hence, an attaining family under the strict inequality $K_{M,n}^* < K_{M,n}^{\text{fix}}$, cf.

Lemma 1, approaches $\mathcal{P}^*$ while remaining outside it for all sufficiently large γ . Hence, the strict inequality arises from the relative rates at which an SNR-dependent codebook family approaches the packing-optimal set $\mathcal{P}^*$.

To make this more precise, along a convergent subsequence, reindexed by γ , write $\mathcal{C}_\gamma \rightarrow \mathcal{C}_\infty \in \mathcal{P}^*$. For an ordered pair $e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)$, define its signed correlation offset from the optimal packing value and compare its single-pair tail scale with B_γ^* :

$$\Delta_{e,\gamma} := \rho_e(\mathcal{C}_\gamma) - \rho_{M,n}^*, \quad (24)$$

$$\frac{b_\gamma(\rho_e(\mathcal{C}_\gamma))}{B_\gamma^*} = \exp\left\{\frac{n\gamma\Delta_{e,\gamma}}{4}\right\} \sqrt{\frac{1 - \rho_{M,n}^*}{1 - \rho_e(\mathcal{C}_\gamma)}}. \quad (25)$$

Because $\rho_e(\mathcal{C}_\gamma) \rightarrow \rho_{M,n}^*$, the square-root factor tends to one. As the limiting-closest-pair result below makes rigorous, for the family under discussion that attains the optimal leading coefficient, if $n\gamma\Delta_{e,\gamma} \rightarrow \lambda_e \in \mathbb{R} \cup \{-\infty\}$, then the contribution of the ordered pair e to $K_{M,n}^*$ is $\exp\{\lambda_e/4\}$, where $\exp\{-\infty\} = 0$. Thus a zero scaled offset gives a unit contribution to $K_{M,n}^*$, whereas divergence to $-\infty$ suppresses the contribution even though the pair is closest in the packing-optimal limit $\mathcal{C}_\infty$.

A revealing mechanism for achieving $K_{M,n}^* < K_{M,n}^{\text{fix}}$, applied in the orthoplex-bound range, is the following. Partition the closest-pair indices in $\mathcal{I}_{\max}(\mathcal{C}_\infty)$ into two classes, denoting indices in the selected class by e and those in the remaining class by f . At finite SNR γ , arrange their correlations so that

$$\begin{aligned} \rho_e(\mathcal{C}_\gamma) &= \rho_{\max}(\mathcal{C}_\gamma) = \rho_{M,n}^* + p_\gamma, & p_\gamma > 0, & \quad p_\gamma \rightarrow 0, & \quad n\gamma p_\gamma \rightarrow 0, \\ \rho_f(\mathcal{C}_\gamma) &= \rho_{M,n}^* - q_{f,\gamma}, & q_{f,\gamma} > 0, & \quad q_{f,\gamma} \rightarrow 0, & \quad n\gamma q_{f,\gamma} \rightarrow \infty. \end{aligned} \quad (26)$$

The first pairs are slightly closer than under an optimal packing, but their deterioration is too small to change the optimal error exponent, and they make unit contributions to the family's leading coefficient. The second pairs are farther apart by a larger vanishing amount than under an optimal packing, so their contributions to the family's leading coefficient vanish, even though they become closest pairs in $\mathcal{C}_\infty$. Thus an asymptotically negligible worsening of selected pairs can create enough geometric freedom to suppress the contributions of other pairs to the family's leading coefficient. This description assumes that the family's leading coefficient exists. More general approach rates may produce fractional pair contributions, so the leading coefficient of an SNR-dependent family, when it exists, need not be an integer-valued ordered closest-pair count. If the family attains the optimal leading coefficient, its leading coefficient equals $K_{M,n}^*$.

If $\mathcal{C}_\infty$ were instead held fixed, every one of its ordered closest-pair indices would contribute one to its fixed-codebook leading coefficient $|\mathcal{I}_{\max}(\mathcal{C}_\infty)|$, which is at least $K_{M,n}^{\text{fix}}$. For an SNR-dependent family that attains the optimal leading coefficient, the pairwise contributions to $K_{M,n}^*$ therefore depend on the family's approach to the limiting packing, not on the limiting packing alone.

III. SOFT-PACKING CHARACTERIZATION OF THE OPTIMAL LEADING COEFFICIENT

The design problem in Subsection II-D first asks whether the normalized SNR-wise optimal ML error has a finite limit and how that limit can be characterized. This section resolves these questions for every fixed finite pair (M, n) satisfying $M \geq 2$ and $\rho_{M,n}^* < 1$. We introduce a Gaussian soft-packing energy, prove that its minimum is asymptotically equivalent to the normalized exact ML optimum, and show that this minimum converges to the optimal leading coefficient $K_{M,n}^*$. Comparison with fixed packing-optimal codebooks then gives $K_{M,n}^* \leq K_{M,n}^{\text{fix}}$.

A. Gaussian Soft-Packing Energy

For every codebook $\mathcal{C} \in \mathcal{M}$, define the Gaussian soft-packing energy as

$$\mathcal{G}_\gamma(\mathcal{C}) = \sum_{e \in \mathcal{E}_M} \exp \left\{ \frac{n\gamma}{4} (\rho_e(\mathcal{C}) - \rho_{M,n}^*) \right\}. \quad (27)$$

Each exponential term in $\mathcal{G}_\gamma(\mathcal{C})$ distinguishes correlation offsets on the $1/(n\gamma)$ scale: an offset $\lambda/(n\gamma)$ produces the finite term $\exp\{\lambda/4\}$ in $\mathcal{G}_\gamma$. The Gaussian soft-packing energy also controls the hard packing objective. Indeed, an ordered pair attaining $\rho_{\max}(\mathcal{C})$ contributes

$$\exp \left\{ \frac{n\gamma}{4} (\rho_{\max}(\mathcal{C}) - \rho_{M,n}^*) \right\}$$

to $\mathcal{G}_\gamma(\mathcal{C})$. Therefore, if $\mathcal{G}_\gamma(\mathcal{C}) \leq L$ for some $L \geq 1$, then

$$0 \leq \rho_{\max}(\mathcal{C}) - \rho_{M,n}^* \leq \frac{4 \log L}{n\gamma}. \quad (28)$$

Thus a bounded Gaussian soft-packing energy $\mathcal{G}_\gamma(\mathcal{C})$ forces the worst correlation into the $1/(n\gamma)$ -scale neighborhood of the optimal packing value $\rho_{M,n}^*$.

For a fixed packing-optimal codebook $\mathcal{C}_0 \in \mathcal{P}^*$, every $e \in \mathcal{I}_{\max}(\mathcal{C}_0)$ has zero correlation offset from $\rho_{M,n}^*$ and thereby contributes one to $\mathcal{G}_\gamma(\mathcal{C}_0)$. Every remaining ordered pair has a fixed negative offset from $\rho_{M,n}^*$; hence its term decays exponentially to zero as $\gamma \rightarrow \infty$. Consequently,

$$\lim_{\gamma \rightarrow \infty} \mathcal{G}_\gamma(\mathcal{C}_0) = |\mathcal{I}_{\max}(\mathcal{C}_0)|. \quad (29)$$

Thus the limiting Gaussian soft-packing energy of a fixed packing-optimal codebook is exactly its number of ordered closest-pair indices. We next compare the Gaussian soft-packing energy with the normalized exact ML error while the codebook is allowed to vary with SNR.

B. Uniform Approximation of the Exact ML Error

The minimum of the Gaussian soft-packing energy over all codebooks $\mathcal{C} \in \mathcal{M}$ at a fixed SNR γ is

$$V_\gamma(M, n) = \min_{\mathcal{C} \in \mathcal{M}} \mathcal{G}_\gamma(\mathcal{C}). \quad (30)$$

For each fixed γ , the map $\mathcal{C} \mapsto \mathcal{G}_\gamma(\mathcal{C})$ is continuous on the compact codebook space $\mathcal{M}$; hence the minimum is attained. The quantity $V_\gamma(M, n)$ plays a crucial role in characterizing the optimal leading coefficient.

Theorem 1 (Uniform Gaussian soft-packing transfer). *Fix finite (M, n) with $M \geq 2$ and $\rho_{M,n}^* < 1$. For every fixed $L \geq 1$, there are constants $C_L < \infty$ and $\gamma_L < \infty$ such that, for every $\gamma \geq \gamma_L$ and every $\mathcal{C} \in \mathcal{M}$ satisfying*

$$\mathcal{G}_\gamma(\mathcal{C}) \leq L, \quad (31)$$

we have

$$\left| \frac{P_e(\mathcal{C}; \gamma)}{B_\gamma^*} - \mathcal{G}_\gamma(\mathcal{C}) \right| \leq \frac{C_L}{n\gamma}. \quad (32)$$

The constants may depend on (M, n) , L , and the positive separation $1 - \rho_{M,n}^$, but not on $\mathcal{C}$ or γ .*

Proof: The proof is given in Appendix A. ■

Theorem 1 shows that, for every fixed $L \geq 1$, the soft-packing energy $\mathcal{G}_\gamma(\mathcal{C})$ uniformly approximates the normalized exact ML error $\frac{P_e(\mathcal{C}; \gamma)}{B_\gamma^*}$ over all codebooks $\mathcal{C} \in \mathcal{M}$ satisfying $\mathcal{G}_\gamma(\mathcal{C}) \leq L$. Equivalently, an SNR-dependent family is packing-tail competitive precisely when $P_e(\mathcal{C}_\gamma; \gamma)/B_\gamma^* = O(1)$. Later on, cf. Lemma 1, we show that every packing-tail-competitive family eventually satisfies such a fixed soft-packing energy bound.

Corollary 1 (Best packing-optimal benchmark). *For every fixed finite (M, n) with $M \geq 2$ and $\rho_{M,n}^* < 1$,*

$$\frac{P_e^{\text{fix}}(M, n; \gamma)}{B_\gamma^*} = K_{M,n}^{\text{fix}} + O((n\gamma)^{-1}). \quad (33)$$

Proof: For every $\mathcal{C}_0 \in \mathcal{P}^*$, all correlation offsets from $\rho_{M,n}^*$ are nonpositive, and hence $\mathcal{G}_\gamma(\mathcal{C}_0) \leq M(M-1)$. Applying Theorem 1 uniformly over $\mathcal{P}^*$ and then taking minima gives

$$\frac{P_e^{\text{fix}}(M, n; \gamma)}{B_\gamma^*} = \min_{\mathcal{C}_0 \in \mathcal{P}^*} \mathcal{G}_\gamma(\mathcal{C}_0) + O((n\gamma)^{-1}).$$

Every $\mathcal{C}_0 \in \mathcal{P}^*$ satisfies $\mathcal{G}_\gamma(\mathcal{C}_0) \geq |\mathcal{I}_{\max}(\mathcal{C}_0)| \geq K_{M,n}^{\text{fix}}$. Choose $\mathcal{C}_0^{\text{fix}} \in \mathcal{P}^*$ with $|\mathcal{I}_{\max}(\mathcal{C}_0^{\text{fix}})| = K_{M,n}^{\text{fix}}$. Its remaining ordered pairs, if any, lie a fixed positive correlation gap below $\rho_{M,n}^*$. Therefore,

$$\mathcal{G}_\gamma(\mathcal{C}_0^{\text{fix}}) = K_{M,n}^{\text{fix}} + O(\exp\{-cn\gamma\})$$

for some $c > 0$. These two bounds prove (33). ■

Corollary 2 (Finite-SNR equivalence of the optimized exact ML error and Gaussian soft-packing energy). *For every fixed finite (M, n) with $M \geq 2$ and $\rho_{M,n}^* < 1$,*

$$\frac{P_e^*(M, n; \gamma)}{B_\gamma^*} = V_\gamma(M, n) + O((n\gamma)^{-1}). \quad (34)$$

Proof: The proof is given in Appendix A. ■

The corollary shows that minimizing the normalized exact ML error and minimizing the soft-packing energy, see (30), produce asymptotically identical optimal normalized values. This result compares the two optimized values, $P_e^*(M, n; \gamma)/B_\gamma^*$ and $V_\gamma(M, n)$, at each sufficiently large finite SNR. It does not yet establish the existence of an optimal leading coefficient, because it does not show that $V_\gamma(M, n)$ converges as $\gamma \rightarrow \infty$.

C. Existence and Characterization of the Optimal Leading Coefficient

It remains to prove that $V_\gamma(M, n)$ has a high-SNR limit. The next theorem establishes this convergence and transfers the resulting limit to the minimum error probability using Corollary 2.

Theorem 2 (Existence and Gaussian soft-packing energy characterization of the optimal leading coefficient). *For every fixed finite (M, n) with $M \geq 2$ and $\rho_{M,n}^* < 1$, the finite limits*

$$\begin{aligned} K_{M,n}^* &:= \lim_{\gamma \rightarrow \infty} V_\gamma(M, n) \\ &= \lim_{\gamma \rightarrow \infty} \frac{P_e^*(M, n; \gamma)}{B_\gamma^*} \end{aligned} \quad (35)$$

exist and agree. The common value, $K_{M,n}^$, satisfies*

$$2 \leq K_{M,n}^* \leq M(M-1). \quad (36)$$

The globally optimal ML error obeys

$$P_e^*(M, n; \gamma) = K_{M,n}^* B_\gamma^* + o(B_\gamma^*). \quad (37)$$

Moreover, for every fixed packing-optimal codebook $\mathcal{C}_0 \in \mathcal{P}^*$,

$$K_{M,n}^* \leq |\mathcal{I}_{\max}(\mathcal{C}_0)|. \quad (38)$$

Proof: The proof is given in Appendix A. ■

The theorem shows that both $P_e^*(M, n; \gamma)/B_\gamma^*$ and $V_\gamma(M, n)$ converge to $K_{M,n}^*$, and that this common limit is at most $|\mathcal{I}_{\max}(\mathcal{C}_0)|$ for every fixed packing-optimal codebook $\mathcal{C}_0$. The scale B_γ^* contains the packing-determined exponential decay and its Gaussian prefactor, while $K_{M,n}^*$ is the remaining optimal leading coefficient.

The theorem also proves existence of the optimized leading coefficient. Specifically, both $P_e^*(M, n; \gamma)/B_\gamma^*$ and $V_\gamma(M, n)$ converge to $K_{M,n}^*$. Consequently, every selection of exact finite-SNR minimizers

$$\mathcal{C}_\gamma^* \in \arg \min_{\mathcal{C} \in \mathcal{M}} P_e(\mathcal{C}; \gamma)$$

has leading coefficient $K_{M,n}^*$, even if the selected optimizing codebooks do not themselves converge as a family. Indeed,

$$\frac{P_e(\mathcal{C}_\gamma^*; \gamma)}{B_\gamma^*} = \frac{P_e^*(M, n; \gamma)}{B_\gamma^*} \longrightarrow K_{M,n}^*.$$

The same characterization also identifies a broad class of families attaining the optimal leading coefficient. Exact minimization of the Gaussian soft-packing energy at every SNR is unnecessary; a globally vanishing additive objective gap is sufficient, as shown in the following.

Corollary 3 (Near-minimizers of the Gaussian soft-packing energy attain the optimal leading coefficient). *Let $\{\widehat{\mathcal{C}}_\gamma\} \subset \mathcal{M}$ and suppose that, for some $\varepsilon_\gamma \geq 0$ satisfying $\varepsilon_\gamma \rightarrow 0$,*

$$0 \leq \mathcal{G}_\gamma(\widehat{\mathcal{C}}_\gamma) - V_\gamma(M, n) \leq \varepsilon_\gamma. \quad (39)$$

Then

$$\begin{aligned} \mathcal{G}_\gamma(\widehat{\mathcal{C}}_\gamma) &\longrightarrow K_{M,n}^*, \\ \frac{P_e(\widehat{\mathcal{C}}_\gamma; \gamma)}{B_\gamma^*} &\longrightarrow K_{M,n}^*, \end{aligned} \quad (40)$$

and

$$\frac{P_e(\widehat{\mathcal{C}}_\gamma; \gamma)}{P_e^*(M, n; \gamma)} \longrightarrow 1. \quad (41)$$

Proof: The proof is given in Appendix A. ■

Thus, any full SNR-indexed family $\{\hat{\mathcal{C}}_\gamma\}_\gamma$ whose Gaussian soft-packing energy $\mathcal{G}_\gamma(\hat{\mathcal{C}}_\gamma)$ is within an additive $o(1)$ of the global minimum $V_\gamma(M, n)$ as $\gamma \rightarrow \infty$ attains the optimal leading coefficient $K_{M,n}^*$, even if its codebooks do not minimize the exact ML error at any finite SNR.

D. Comparison With the Best Packing-Optimal Benchmark

The fixed-codebook comparison can be written directly via the Gaussian soft-packing energy function as

$$\begin{aligned} K_{M,n}^* &= \lim_{\gamma \rightarrow \infty} \min_{\mathcal{C} \in \mathcal{M}} \mathcal{G}_\gamma(\mathcal{C}) \\ &\leq \min_{\mathcal{C}_0 \in \mathcal{P}^*} \lim_{\gamma \rightarrow \infty} \mathcal{G}_\gamma(\mathcal{C}_0) = K_{M,n}^{\text{fix}}. \end{aligned} \tag{42}$$

On the left, the codebook is optimized separately at every SNR before the high-SNR limit is taken. On the right, a packing-optimal codebook is first held fixed, its limiting number of ordered closest-pair indices is obtained, and that count is then minimized over $\mathcal{P}^*$. The first procedure retains the path-dependent pair contributions to the optimal leading coefficient that can arise under SNR-wise codebook optimization; the second assigns one unit to the fixed codebook's leading coefficient for every ordered closest-pair index of that codebook.

In fact, (20) gives

$$P_e^*(M, n; \gamma) \leq P_e^{\text{fix}}(M, n; \gamma) \quad \text{for every } \gamma.$$

Theorem 2 and Corollary 1 then give

$$\begin{aligned} \frac{P_e^*(M, n; \gamma)}{P_e^{\text{fix}}(M, n; \gamma)} &\longrightarrow \frac{K_{M,n}^*}{K_{M,n}^{\text{fix}}}, \\ \frac{P_e^{\text{fix}}(M, n; \gamma) - P_e^*(M, n; \gamma)}{B_\gamma^*} &\longrightarrow K_{M,n}^{\text{fix}} - K_{M,n}^*. \end{aligned} \tag{43}$$

Therefore, under the strict coefficient inequality, $K_{M,n}^* < K_{M,n}^{\text{fix}}$, the packing-optimal restriction retains a constant multiplicative disadvantage, and the exact SNR-wise optimal error is strictly smaller for every sufficiently large SNR. If the coefficients are equal, $K_{M,n}^* = K_{M,n}^{\text{fix}}$, the best fixed packing-optimal codebook is leading-order optimal, although lower-order differences may remain.

The limit characterization determines the optimal leading coefficient but does not yet describe the geometry of the codebook families that can approach it. The next section supplies that structural description and derives a global variational characterization.

IV. LOCALIZATION AND SNR-DEPENDENT CLOSEST-PAIR CONTRIBUTIONS

The preceding section identifies the optimal leading coefficient as the high-SNR limit of the minimum Gaussian soft-packing energy, $K_{M,n}^* = \lim_{\gamma \rightarrow \infty} V_\gamma(M, n)$. This section determines which ordered pairs can have nonvanishing normalized contributions along a family competitive at the packing-tail scale and how those contributions are determined. The section concludes by minimizing these contributions over all packing-optimal limiting codebooks and all jointly feasible scaled closest-pair offsets.

A. Localization at the Packing-Tail Scale

Here and below, $\text{dist}(\mathcal{C}, \mathcal{P}^*)$ is induced by the product Euclidean metric on the labeled tuples in $\mathcal{M}$.

Lemma 1 (Localization at the packing-tail scale). *Let $\{\mathcal{C}_\gamma\} \subset \mathcal{M}$ satisfy*

$$\frac{P_e(\mathcal{C}_\gamma; \gamma)}{B_\gamma^*} = O(1). \quad (44)$$

Then

$$\rho_{\max}(\mathcal{C}_\gamma) - \rho_{M,n}^* = O((n\gamma)^{-1}), \quad (45)$$

$$\mathcal{G}_\gamma(\mathcal{C}_\gamma) = O(1), \quad \text{dist}(\mathcal{C}_\gamma, \mathcal{P}^*) \rightarrow 0. \quad (46)$$

If the family additionally attains the optimal leading coefficient in the sense of (22) and $K_{M,n}^ < K_{M,n}^{\text{fix}}$, then*

$$\mathcal{C}_\gamma \notin \mathcal{P}^* \quad \text{for all sufficiently large } \gamma. \quad (47)$$

Proof: The proof is given in Appendix B. ■

Lemma 1 shows that any packing-tail-competitive codebook family must enter an increasingly small neighborhood of the packing-optimal set. Indeed, define the excess maximal correlation as $\delta_\gamma = \rho_{\max}(\mathcal{C}_\gamma) - \rho_{M,n}^*$. If $\delta_\gamma > 0$, then the single-pair Gaussian-tail scale, normalized by B_γ^* , grows essentially as $\exp\left\{\frac{n\gamma\delta_\gamma}{4}\right\}$. Consequently, bounded normalized error is possible only if (45) holds. Thus, the maximal correlation must be within $O((n\gamma)^{-1})$ of $\rho_{M,n}^*$, which is the critical correlation scale governing pairwise contributions to a family's leading coefficient, whenever that coefficient exists. The lemma also establishes the geometric localization $\text{dist}(\mathcal{C}_\gamma, \mathcal{P}^*) \rightarrow 0$. Hence, the family may continue to rotate, relabel its codewords, or move among different packing-optimal configurations while its distance from $\mathcal{P}^*$ tends to zero as the SNR grows.

The lemma also reveals the geometric mechanism by which the optimal leading coefficient can be smaller than the best fixed-codebook leading coefficient. If $K_{M,n}^* < K_{M,n}^{\text{fix}}$, then a family attaining the optimal leading coefficient must lie outside $\mathcal{P}^*$ for every sufficiently large γ . Instead, it approaches $\mathcal{P}^*$ from outside. Such a family accepts a vanishing increase in its maximal correlation, confined to the $1/(n\gamma)$ scale, in order to reduce or suppress the contributions to $K_{M,n}^*$ made by other pairs that are closest in a packing-optimal subsequential limit. Therefore, such a family may consist of codebooks that are slightly worse as packings at every sufficiently large finite SNR and nevertheless have a strictly smaller leading coefficient than every fixed packing-optimal codebook.

B. Reduction to Closest Pairs of a Limiting Packing

Consider a convergent subsequence, reindexed by γ , for which $\mathcal{C}_\gamma \rightarrow \mathcal{C}_\infty \in \mathcal{P}^*$. Thus $\mathcal{C}_\infty$ is a packing-optimal subsequential limit of the family. For each $e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)$, use the signed correlation offset from (24),

$$\Delta_{e,\gamma} = \rho_e(\mathcal{C}_\gamma) - \rho_{M,n}^*.$$

A positive offset worsens that ordered pair relative to the packing value, whereas a negative offset improves it.

Theorem 3 (Reduction to closest pairs of a limiting packing). *Let $\mathcal{C}_\gamma \rightarrow \mathcal{C}_\infty \in \mathcal{P}^*$, and suppose that, for some finite $L \geq 1$, $\mathcal{G}_\gamma(\mathcal{C}_\gamma) \leq L$ for all sufficiently large γ . Then, for all sufficiently large γ ,*

$$0 \leq \rho_{\max}(\mathcal{C}_\gamma) - \rho_{M,n}^* = \max_{e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)} \Delta_{e,\gamma} \leq \frac{4 \log L}{n\gamma}.$$

Moreover, for some $c > 0$,

$$\mathcal{G}_\gamma(\mathcal{C}_\gamma) = \sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)} \exp \left\{ \frac{n\gamma \Delta_{e,\gamma}}{4} \right\} + O(\exp\{-cn\gamma\}), \quad (48)$$

$$\begin{aligned} \frac{P_e(\mathcal{C}_\gamma; \gamma)}{B_\gamma^*} &= \sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)} \exp \left\{ \frac{n\gamma \Delta_{e,\gamma}}{4} \right\} \\ &\quad + O((n\gamma)^{-1}). \end{aligned} \quad (49)$$

Proof: The proof is given in Appendix B. ■

Theorem 3 shows that, for a packing-tail-competitive codebook family that converges to an optimal packing, not all ordered codeword pairs can contribute to the normalized error

$P_e(\mathcal{C}_\gamma; \gamma)/B_\gamma^*$. The limiting packing restricts the ordered pairs that can contribute to the normalized error to those that are closest pairs in the limiting optimal packing. Indeed, if an ordered pair e is not in $\mathcal{I}_{\max}(\mathcal{C}_\infty)$, then, for some fixed $\eta_e > 0$, $\rho_e(\mathcal{C}_\infty) = \rho_{M,n}^* - \eta_e$. Because $\mathcal{C}_\gamma \rightarrow \mathcal{C}_\infty$, this pair retains a strict correlation gap for all sufficiently large γ . Therefore, its Gaussian soft-packing energy term is bounded by an exponentially vanishing quantity of the form $\exp\{-n\gamma\eta_e/8\}$, and its contribution to the normalized error vanishes. For an ordered pair that is closest in the limiting packing, i.e., $e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)$, by contrast, $\rho_e(\mathcal{C}_\infty) = \rho_{M,n}^*$, so there is no fixed correlation gap that makes its contribution to the normalized error vanish. Instead, its finite-SNR displacement $\Delta_{e,\gamma} = \rho_e(\mathcal{C}_\gamma) - \rho_{M,n}^*$ enters the normalized error through the term $\exp\{n\gamma\Delta_{e,\gamma}/4\}$. Consequently, the theorem reduces the normalized error to (49).

Ordinary geometric convergence determines the set of potentially relevant closest pairs but does not determine each contribution in this sum. It gives only $\Delta_{e,\gamma} \rightarrow 0$, whereas the limiting normalized error along a subsequence depends on the finer quantities $n\gamma\Delta_{e,\gamma}$. Thus, different SNR-dependent approaches to the same optimal packing can yield different subsequential limits of the normalized error. The limiting codebook is the common destination, whereas the approach to that destination on the $1/(n\gamma)$ correlation scale determines each limiting closest pair's normalized contribution along the selected subsequence. The following corollary makes these contributions explicit when the scaled offsets have limits.

Corollary 4 (Subsequential closest-pair contribution formula). *Under the assumptions of Theorem 3, since $\mathcal{I}_{\max}(\mathcal{C}_\infty)$ is finite and a bounded Gaussian soft-packing energy precludes any scaled offset from diverging to $+\infty$, there exists a further subsequence, without changing notation, such that*

$$n\gamma\Delta_{e,\gamma} \longrightarrow \lambda_e, \quad e \in \mathcal{I}_{\max}(\mathcal{C}_\infty), \quad (50)$$

simultaneously for some $\lambda_e \in \mathbb{R} \cup \{-\infty\}$, where $\exp\{-\infty\} = 0$. Along this subsequence,

$$\mathcal{G}_\gamma(\mathcal{C}_\gamma) \longrightarrow \sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)} \exp\{\lambda_e/4\}, \quad (51)$$

$$\frac{P_e(\mathcal{C}_\gamma; \gamma)}{B_\gamma^*} \longrightarrow \sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)} \exp\{\lambda_e/4\}. \quad (52)$$

If the family attains the optimal leading coefficient in the sense of (22), then

$$K_{M,n}^* = \sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)} \exp\{\lambda_e/4\}.$$

For the constant family $\mathcal{C}_\gamma \equiv \mathcal{C}_\infty$, every $e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)$ has $\Delta_{e,\gamma} = 0$ and therefore contributes one to both limiting sums. Hence,

$$\lim_{\gamma \rightarrow \infty} \mathcal{G}_\gamma(\mathcal{C}_\infty) = \lim_{\gamma \rightarrow \infty} \frac{P_e(\mathcal{C}_\infty; \gamma)}{B_\gamma^*} = |\mathcal{I}_{\max}(\mathcal{C}_\infty)| \geq K_{M,n}^{\text{fix}}.$$

Thus, a fixed packing-optimal codebook has a leading coefficient equal to its number of ordered closest-pair indices; this coefficient equals $K_{M,n}^*$ precisely when that constant family attains the optimal leading coefficient.

Proof: Equations (48) and (49) give the two displayed limits. Bounded Gaussian soft-packing energy gives $n\gamma\Delta_{e,\gamma} \leq 4\log L$ for every $e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)$. The extended interval $[-\infty, 4\log L]$ is compact, and the set $\mathcal{I}_{\max}(\mathcal{C}_\infty)$ is finite, so a common convergent subsequence exists. The attainment conclusion follows from (22). The constant-family conclusion follows from $\Delta_{e,\gamma} \equiv 0$ and (16). ■

Corollary 4 converts the scaled correlation offset of each limiting closest pair into its contribution $\exp\{\lambda_e/4\}$ to the limiting normalized error along that subsequence. If the full family has a leading coefficient, these contributions sum to that coefficient. If the family attains the optimal leading coefficient, their sum is $K_{M,n}^*$.

A fixed packing-optimal codebook provides the simplest special case. If $\mathcal{C}_\gamma = \mathcal{C}_\infty$, then every ordered closest-pair index has zero offset, so that $\lambda_e = 0$ and $\exp\{\lambda_e/4\} = 1$. Its leading coefficient is therefore the number of ordered closest-pair indices. For an SNR-dependent family, however, ordinary convergence still makes every such offset tend to zero, but multiplication by $n\gamma$ reveals differences that remain visible after normalization by B_γ^* . In particular, if $n\gamma\Delta_{e,\gamma} \rightarrow \lambda_e$, the limiting pair contribution has one of the following forms:

scaled-offset behavior	limiting pair contribution
$\lambda_e = -\infty$	$\exp\{\lambda_e/4\} = 0,$
$-\infty < \lambda_e < 0$	$0 < \exp\{\lambda_e/4\} < 1,$
$\lambda_e = 0$	$\exp\{\lambda_e/4\} = 1,$
$\lambda_e > 0$	$\exp\{\lambda_e/4\} > 1.$

For a family attaining the optimal leading coefficient, these are the corresponding contributions to $K_{M,n}^*$. Consequently, a pair that is closest in the limiting codebook may have a zero, fractional, unit, or greater-than-unit contribution, depending on the $1/(n\gamma)$ -scale path by which the codebook approaches the limiting packing.

C. Global Variational Characterization by Scaled Closest-Pair Offsets

Corollary 4 evaluates one specified SNR-dependent family. To characterize the global optimum, the feasible subsequential scaled offsets must be considered jointly over every packing-optimal limit.

For $\mathcal{C}_0 \in \mathcal{P}^*$, let $\mathfrak{L}(\mathcal{C}_0)$ denote the set of all vectors

$$\boldsymbol{\lambda} = (\lambda_e)_{e \in \mathcal{I}_{\max}(\mathcal{C}_0)} \in (\mathbb{R} \cup \{-\infty\})^{\mathcal{I}_{\max}(\mathcal{C}_0)}$$

for which there exist SNRs $\gamma_j \rightarrow \infty$ and codebooks $\mathcal{C}_j \in \mathcal{M}$ satisfying

$$\begin{aligned} \mathcal{C}_j &\longrightarrow \mathcal{C}_0, \quad \sup_j \mathcal{G}_{\gamma_j}(\mathcal{C}_j) < \infty, \\ n\gamma_j(\rho_e(\mathcal{C}_j) - \rho_{M,n}^*) &\longrightarrow \lambda_e, \quad e \in \mathcal{I}_{\max}(\mathcal{C}_0). \end{aligned} \tag{53}$$

We call $\boldsymbol{\lambda}$ admissible scaled closest-pair-offset data at $\mathcal{C}_0$. The bounded Gaussian soft-packing energy condition excludes $+\infty$ as a component limit, whereas $-\infty$ is retained and gives the limiting pair weight $\exp\{-\infty\} = 0$. The constant sequence $\mathcal{C}_j \equiv \mathcal{C}_0$ shows that $\mathfrak{L}(\mathcal{C}_0)$ is nonempty.

Proposition 2 (Global variational characterization by scaled closest-pair offsets). *For every fixed finite (M, n) with $M \geq 2$ and $\rho_{M,n}^* < 1$,*

$$K_{M,n}^* = \min_{\substack{\mathcal{C}_0 \in \mathcal{P}^* \\ \boldsymbol{\lambda} \in \mathfrak{L}(\mathcal{C}_0)}} \sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_0)} \exp\{\lambda_e/4\}. \tag{54}$$

The minimum in (54) is attained. Moreover, every sequence witnessing a minimizing pair $(\mathcal{C}_0, \boldsymbol{\lambda})$ satisfies, along its SNR sequence,

$$\mathcal{G}_{\gamma_j}(\mathcal{C}_j) \longrightarrow K_{M,n}^*, \quad \frac{P_e(\mathcal{C}_j; \gamma_j)}{B_{\gamma_j}^*} \longrightarrow K_{M,n}^*, \quad \frac{P_e(\mathcal{C}_j; \gamma_j)}{P_e^*(M, n; \gamma_j)} \longrightarrow 1.$$

Proof: The proof is given in Appendix B. ■

The proposition minimizes over every packing-optimal limiting codebook and every jointly feasible set of scaled offsets. The limiting packing fixes the ordered pairs in the sum, while the offsets determine their limiting weights. A constant packing has $\lambda_e = 0$ at every ordered closest-pair index and therefore recovers the ordinary closest-pair count. The joint-feasibility condition is essential: the scaled offsets must arise simultaneously from one feasible codebook sequence and therefore cannot be optimized independently. The characterization of $K_{M,n}^*$ is exact but generally implicit because the admissible sets $\mathfrak{L}(\mathcal{C}_0)$ need not admit tractable finite-dimensional descriptions.

V. LEADING-COEFFICIENT EXTRACTION BY MATCHING BOUNDS

The preceding sections establish the existence of $K_{M,n}^*$ and characterize the closest-pair contributions that can produce it. To determine $K_{M,n}^*$ in closed form, the applications below use a direct matching-bounds argument. Specifically, since

$$K_{M,n}^* = \lim_{\gamma \rightarrow \infty} V_\gamma(M, n),$$

it is enough to identify a candidate value K , prove the global lower bound $\liminf_{\gamma \rightarrow \infty} V_\gamma(M, n) \geq K$, and construct a full SNR-indexed family whose Gaussian soft-packing energy has limiting upper bound at most K . The following corollary then identifies K as the optimal leading coefficient, transfers the equality to the exact SNR-wise optimized ML error, and certifies the constructed family.

Corollary 5 (Optimal leading-coefficient extraction by matching bounds). *Fix a finite pair (M, n) with $M \geq 2$ and $\rho_{M,n}^* < 1$, and let $K \in [2, M(M-1)]$. Suppose that*

$$\liminf_{\gamma \rightarrow \infty} V_\gamma(M, n) \geq K, \tag{55}$$

and that there is an SNR-indexed family, defined for every sufficiently large γ , $\{\mathcal{C}_\gamma\} \subset \mathcal{M}$ satisfying

$$\limsup_{\gamma \rightarrow \infty} \mathcal{G}_\gamma(\mathcal{C}_\gamma) \leq K. \tag{56}$$

Then

$$K_{M,n}^* = K, \quad P_e^*(M, n; \gamma) = B_\gamma^*(K + o(1)), \tag{57}$$

and the constructed family satisfies

$$P_e(\mathcal{C}_\gamma; \gamma) = B_\gamma^*(K + o(1)), \quad \frac{P_e(\mathcal{C}_\gamma; \gamma)}{P_e^*(M, n; \gamma)} \rightarrow 1. \tag{58}$$

Proof: Since $V_\gamma(M, n) \leq \mathcal{G}_\gamma(\mathcal{C}_\gamma)$,

$$\begin{aligned} K &\leq \liminf_{\gamma \rightarrow \infty} V_\gamma(M, n) \\ &\leq \limsup_{\gamma \rightarrow \infty} V_\gamma(M, n) \\ &\leq \limsup_{\gamma \rightarrow \infty} \mathcal{G}_\gamma(\mathcal{C}_\gamma) \\ &\leq K. \end{aligned}$$

Thus $V_\gamma(M, n) \rightarrow K$, and Theorem 2 gives $K_{M,n}^* = K$ and the optimal leading-order expansion. Moreover, $V_\gamma(M, n) \leq \mathcal{G}_\gamma(\mathcal{C}_\gamma)$ gives $\liminf_{\gamma \rightarrow \infty} \mathcal{G}_\gamma(\mathcal{C}_\gamma) \geq K$, while (56) gives the reverse upper limit. Hence $\mathcal{G}_\gamma(\mathcal{C}_\gamma) \rightarrow K$. This convergence places the family in a fixed bounded-energy sublevel of $\mathcal{G}_\gamma$, so Theorem 1 yields $P_e(\mathcal{C}_\gamma; \gamma)/B_\gamma^* = K + o(1)$. Together with the optimal leading-order expansion and $K \geq 2$, this also gives the relative-error limit in (58). ■

The lower bound in (55) is the converse: it controls every SNR-dependent codebook family through the global minimum $V_\gamma(M, n)$. The upper bound in (56) is achievability: one feasible family reaches the same value. Once the two bounds match, Corollary 5 and the uniform transfer in Theorem 1 give both the exact SNR-wise optimized ML expansion and attainment of the optimal leading coefficient.

Accordingly, determining the optimal leading coefficient $K_{M,n}^*$ by matching bounds consists of three steps:

- 1) Determine the optimal packing correlation $\rho_{M,n}^*$, which fixes the universal tail scale B_γ^* , and identify a candidate value K .
- 2) Prove the global converse

$$\liminf_{\gamma \rightarrow \infty} V_\gamma(M, n) \geq K.$$

This may follow from a direct inequality valid for every feasible codebook or from localization followed by a problem-specific geometric argument.

- 3) Construct a feasible codebook family, defined for every sufficiently large γ , such that

$$\limsup_{\gamma \rightarrow \infty} \mathcal{G}_\gamma(\mathcal{C}_\gamma) \leq K.$$

Corollary 5 then proves $K_{M,n}^* = K$, the exact SNR-wise optimized ML expansion, and relative-error attainment of the constructed family.

The next two sections apply this three-step matching-bounds framework to the simplex and orthoplex-bound ranges.

VI. APPLICATION TO THE SIMPLEX CODES: EXACT OPTIMAL LEADING COEFFICIENTS

This section applies the matching-bounds framework to $2 \leq M \leq n + 1$. The simplex bound in [4] fixes the packing scale, and a global Jensen inequality solves the soft-packing energy optimization at every positive SNR. The resulting optimal leading coefficient is determined exactly, is attained by a fixed regular simplex, and cannot be improved by SNR-dependent motion.

Fix integers $n \geq 1$ and $2 \leq M \leq n + 1$, and let $\mathcal{C}_{\text{simp}}$ be an embedded regular $(M - 1)$ -simplex in $\mathbb{S}^{n-1}$. This bound gives

$$\rho_{M,n}^* = -\frac{1}{M-1}, \quad (59)$$

with equality only for an embedded regular simplex, up to an orthogonal transformation and relabeling [4]. Substitution in (14) fixes the packing-tail scale as

$$B_\gamma^* = \frac{\exp\{-n\gamma M/[4(M-1)]\}}{M\sqrt{\pi n\gamma M/(M-1)}}. \quad (60)$$

Here the finite-SNR Gaussian soft-packing energy optimization can be solved exactly. The following proposition gives the global minimum and identifies all minimizers.

Proposition 3 (Exact soft-packing energy minimum in the simplex range). *For all integers $n \geq 1$ and $2 \leq M \leq n + 1$, and every $\gamma > 0$,*

$$V_\gamma(M, n) = \mathcal{G}_\gamma(\mathcal{C}_{\text{simp}}) = M(M-1). \quad (61)$$

Moreover, the minimizers are exactly the embedded regular $(M - 1)$ -simplices, up to orthogonal transformations and relabeling.

Proof: The proof is given in Appendix C. ■

The proposition is stronger than a high-SNR statement: an embedded regular simplex globally minimizes the Gaussian soft-packing energy for every $\gamma > 0$. Consequently, no finite-SNR deformation, including one that varies with γ , can lower the Gaussian soft-packing energy minimum below the simplex value.

Corollary 6 (Optimal leading coefficient in the simplex range). *For every fixed pair (M, n) satisfying $2 \leq M \leq n + 1$,*

$$K_{M,n}^* = K_{M,n}^{\text{fix}} = M(M-1), \quad (62)$$

and

$$\begin{aligned} P_e^*(M, n; \gamma) &= B_\gamma^*(M(M-1) + o(1)) \\ &= \frac{(M-1)^{3/2}}{\sqrt{\pi n\gamma M}} \exp\left\{-\frac{n\gamma M}{4(M-1)}\right\} (1 + o(1)). \end{aligned} \quad (63)$$

The constant family $\mathcal{C}_\gamma \equiv \mathcal{C}_{\text{simp}}$ attains $K_{M,n}^$ and satisfies*

$$\frac{P_e(\mathcal{C}_{\text{simp}}; \gamma)}{P_e^*(M, n; \gamma)} \longrightarrow 1. \quad (64)$$

Proof: Proposition 3 gives the global lower bound (55) and, through the constant simplex family, the matching upper bound (56), both with $K = M(M - 1)$. Corollary 5 gives $K_{M,n}^* = M(M - 1)$ and attainment of the optimal leading coefficient; substitution of (60) gives (63). Packing uniqueness and the fact that every ordered pair of distinct simplex vertices is a closest-pair index give $K_{M,n}^{\text{fix}} = M(M - 1)$, completing (62). The relative-error assertion then follows from attainment of the optimal leading coefficient. ■

Corollary 6 completely resolves the SNR-wise optimized high-SNR problem in the simplex range. The optimal and best fixed-codebook leading coefficients coincide, and the constant simplex family attains the optimal leading coefficient; SNR-dependent motion therefore gives no leading-order gain. Proposition 3 establishes exact finite-SNR optimality for the Gaussian soft-packing energy, while the conclusion for the exact ML error follows asymptotically through uniform transfer. Independently, the recent resolution of the Weak Simplex Conjecture in [14, Cors. 2.5–2.6] proves the stronger statement that an embedded regular $(M - 1)$ -simplex minimizes the exact average ML error at every positive SNR in this entire cardinality range. That external result is consistent with, but is not needed for, the leading-coefficient derivation above.

VII. APPLICATION TO THE ORTHOPLEX-BOUND RANGE

Fix $n \geq 2$ and parameterize the orthoplex-bound range by

$$M = n + k, \quad 2 \leq k \leq n, \quad (65)$$

so that $n + 2 \leq M \leq 2n$. This section first determines the packing scale and the best fixed-codebook leading coefficient $K_{n+k,n}^{\text{fix}}$. It then constructs a strictly improving SNR-dependent family for every $2 \leq k < n$, obtains an achievability bound $K_{n+k,n}^* \leq 4k(k - 1)$, determines $K_{n+k,n}^*$ exactly at $k = 2$ and $k = n$, and isolates the remaining intermediate equality as a conjecture.

A. Packing Scale and Best Fixed-Codebook Leading Coefficient

The orthoplex bound in [4] gives $\rho_{M,n}^* \geq 0$ whenever $M > n + 1$. Conversely, since $M = n + k \leq 2n$, any M vertices of the cross-polytope $\{\pm e_1, \dots, \pm e_n\}$ have all off-diagonal correlations in $\{-1, 0\}$. Hence, throughout the range,

$$\rho_{n+k,n}^* = 0. \quad (66)$$

Substitution in (14) gives the packing-tail scale

$$B_\gamma^* = \frac{\exp\{-n\gamma/4\}}{(n + k)\sqrt{n\gamma}\sqrt{\pi}}. \quad (67)$$

Proposition 4 (Best fixed-codebook leading coefficient in the orthoplex-bound range). *For every fixed $n \geq 2$ and $2 \leq k \leq n$,*

$$K_{n+k,n}^{\text{fix}} = 4n(k-1). \quad (68)$$

Equality is attained by the orthogonal union of $k-1$ antipodal pairs and a regular $(n-k+1)$ -simplex, the latter having $n-k+2$ vertices in the remaining $(n-k+1)$ -dimensional subspace.

Proof: The proof is given in Appendix C. ■

Proposition 4 determines the smallest leading coefficient obtainable by holding one packing-optimal codebook fixed. The stated orthogonal union attains the minimum ordered closest-pair-index count $4n(k-1)$. This is the benchmark for measuring the benefit of SNR-wise codebook optimization, developed below.

B. An SNR-Dependent Construction for $2 \leq k < n$

For every fixed $n \geq 3$ and $2 \leq k < n$, put

$$d = n - k, \quad q_\gamma = \frac{\log(n\gamma)}{n\gamma}. \quad (69)$$

The closest-pair contribution formula in Corollary 4 suggests this choice because it separates the two scales

$$\frac{1}{n\gamma} \ll q_\gamma \ll \frac{1}{\sqrt{n\gamma}}.$$

Thus correlations lowered by order q_γ produce zero limiting weights in the Gaussian soft-packing energy, whereas correlations worsened by order q_γ^2 produce unit limiting weights.

For all sufficiently large γ , choose unit vectors $\mathbf{a}_1, \dots, \mathbf{a}_d$ in a d -dimensional subspace $\mathcal{U} \subset \mathbb{R}^n$, continuously in q_γ from an orthonormal basis at $q_\gamma = 0$, such that

$$\mathbf{a}_i^\top \mathbf{a}_j = -q_\gamma, \quad i \neq j. \quad (70)$$

Define

$$\begin{aligned} \mathbf{v}_\gamma &= -\frac{q_\gamma}{1 - (d-1)q_\gamma} \sum_{j=1}^d \mathbf{a}_j, \\ p_\gamma &= \|\mathbf{v}_\gamma\|^2 = \frac{dq_\gamma^2}{1 - (d-1)q_\gamma}, \\ r_\gamma &= \sqrt{1 - p_\gamma}. \end{aligned} \quad (71)$$

Let $\mathbf{f}_1, \dots, \mathbf{f}_k$ be a fixed orthonormal basis of $\mathcal{U}^\perp$, and set

$$\mathcal{C}_\gamma = \{\mathbf{v}_\gamma \pm r_\gamma \mathbf{f}_i : 1 \leq i \leq k\} \cup \{\mathbf{a}_1, \dots, \mathbf{a}_d\}. \quad (72)$$

Proposition 5 (Multiscale achievability in the nonterminal orthoplex-bound range). *For every fixed $n \geq 3$ and $2 \leq k < n$, the vectors in (70) and the codebook in (72) exist for all sufficiently large γ . The codebook is feasible, has $n + k$ points, and converges to a deleted cross-polytope having both signs on k axes and one sign on each of the remaining $d = n - k$ axes. Its maximal correlation is p_γ , and its Gaussian soft-packing energy is*

$$\begin{aligned} \mathcal{G}_\gamma(\mathcal{C}_\gamma) &= 4k(k-1) \exp\{n\gamma p_\gamma/4\} \\ &\quad + 2k \exp\{n\gamma(2p_\gamma - 1)/4\} \\ &\quad + d(d+4k-1) \exp\{-n\gamma q_\gamma/4\}. \end{aligned} \tag{73}$$

Moreover,

$$\begin{aligned} n\gamma q_\gamma &= \log(n\gamma) \longrightarrow \infty, \\ n\gamma p_\gamma &= O\left(\frac{[\log(n\gamma)]^2}{n\gamma}\right) \longrightarrow 0. \end{aligned} \tag{74}$$

Consequently,

$$\mathcal{G}_\gamma(\mathcal{C}_\gamma) \longrightarrow 4k(k-1). \tag{75}$$

In particular,

$$K_{n+k,n}^* \leq 4k(k-1). \tag{76}$$

The family in (72) satisfies

$$P_e(\mathcal{C}_\gamma; \gamma) = B_\gamma^*(4k(k-1) + o(1)). \tag{77}$$

Proof: The proof is given in Appendix C. ■

Proposition 5 gives an achievability family defined for every sufficiently large SNR, rather than only along a subsequence. The construction lowers by order q_γ the correlations of the $d(d+4k-1)$ pairs involving an auxiliary point that are closest in the limiting deleted cross-polytope, so their contributions to the constructed family's leading coefficient vanish because $n\gamma q_\gamma \rightarrow \infty$. The correlations of the $4k(k-1)$ pairs between shifted points on different axes increase only by $p_\gamma = O(q_\gamma^2)$; since $n\gamma p_\gamma \rightarrow 0$, these pairs have limiting weight one. The exact ML error of the constructed family therefore has leading coefficient $4k(k-1)$. The improvement is obtained only because the codebook continues to vary with SNR at the specified two rates.

C. Exact Optimal Leading Coefficients at the Orthoplex-Bound Endpoints

For the orthoplex-range endpoints, we first solve the terminal Gaussian soft-packing energy problem exactly at every positive SNR, which determines the optimal leading coefficient for $k = n$, and then provide a matching lower bound for $k = 2$.

Theorem 4 (Exact terminal endpoint $k = n$). *For every fixed $n \geq 2$ and every $\gamma > 0$, the full cross-polytope*

$$\mathcal{C}_{\text{cross}} = \{\mathbf{e}_1, -\mathbf{e}_1, \dots, \mathbf{e}_n, -\mathbf{e}_n\}$$

is a global minimizer of the Gaussian soft-packing energy, and

$$V_\gamma(2n, n) = 4n(n-1) + 2n \exp\{-n\gamma/4\}. \quad (78)$$

Consequently,

$$K_{2n,n}^* = K_{2n,n}^{\text{fix}} = 4n(n-1), \quad (79)$$

and

$$P_e^*(2n, n; \gamma) = \frac{2(n-1)}{\sqrt{\pi n \gamma}} \exp\{-n\gamma/4\} (1 + o(1)). \quad (80)$$

The fixed full cross-polytope is therefore asymptotically ML-optimal.

Proof: The proof is given in Appendix C. ■

The terminal endpoint behaves like the simplex range. Universal optimality makes the full cross-polytope a soft-packing energy minimizer at every positive SNR. Its $4n(n-1)$ ordered orthogonal pairs contribute one unit each to the limiting Gaussian soft-packing energy, whereas the contributions of the $2n$ ordered antipodal pairs vanish exponentially. Hence $K_{2n,n}^{\text{fix}}$ and $K_{2n,n}^*$ coincide, and SNR-dependent motion cannot improve the leading coefficient at $k = n$. The result also covers the square $n = k = 2$, where the two orthoplex endpoints coincide. The exact finite-SNR assertion concerns the Gaussian soft-packing energy; the exact ML conclusion established here is asymptotic.

To determine the left endpoint of the orthoplex-bound range, the construction must be paired with a global converse valid for every codebook, rather than only for a particular limiting packing.

Proposition 6 (Global signed-balance lower bound). *For every $\mathcal{C} \in (\mathbb{S}^{n-1})^{n+2}$, put $\varrho = \rho_{\max}(\mathcal{C})$. Then, for every $\gamma > 0$,*

$$\mathcal{G}_\gamma(\mathcal{C}) \geq \frac{8}{(1 + \varrho)^2}. \quad (81)$$

Consequently,

$$\liminf_{\gamma \rightarrow \infty} V_\gamma(n+2, n) \geq 8. \quad (82)$$

Proof: The proof is given in Appendix C. ■

Unlike a bound based on a fixed closest-pair count, Proposition 6 applies globally to every $(n+2, n)$ codebook, including non-packing-optimal codebooks and codebooks selected from SNR-dependent families. Every packing-tail-competitive family has $\rho_{\max}(\mathcal{C}_\gamma) \rightarrow 0$, so the finite-SNR inequality forces the liminf of its Gaussian soft-packing energy to be at least 8. It therefore supplies exactly the converse needed to certify $K_{n+2,n}^*$.

Theorem 5 (Exact optimal leading coefficient at the left endpoint $k = 2$). *For every fixed $n \geq 2$, as $\gamma \rightarrow \infty$,*

$$V_\gamma(n+2, n) = 8 + o(1), \quad (83)$$

and

$$K_{n+2,n}^* = 8. \quad (84)$$

The exact SNR-wise optimized ML error satisfies

$$P_e^*(n+2, n; \gamma) = \frac{\exp\{-n\gamma/4\}}{(n+2)\sqrt{n\gamma}\sqrt{\pi}} (8 + o(1)). \quad (85)$$

Moreover, a family attaining the optimal leading coefficient is given by the fixed square when $n = 2$ and by the SNR-dependent codebook in (72), specialized to $k = 2$, when $n \geq 3$. In either case,

$$P_e(\mathcal{C}_\gamma; \gamma) = B_\gamma^*(8 + o(1)), \quad \frac{P_e(\mathcal{C}_\gamma; \gamma)}{P_e^*(n+2, n; \gamma)} \rightarrow 1. \quad (86)$$

Proof: Proposition 6 gives $\liminf_{\gamma \rightarrow \infty} V_\gamma(n+2, n) \geq 8$. Let $\mathcal{C}_\gamma$ be the fixed square when $n = 2$ and the family in (72), specialized to $k = 2$, when $n \geq 3$. In the first case, Theorem 4 gives $\mathcal{G}_\gamma(\mathcal{C}_\gamma) = V_\gamma(4, 2) = 8 + 4e^{-\gamma/2}$. In the second case, (75) gives $\mathcal{G}_\gamma(\mathcal{C}_\gamma) \rightarrow 8$. Since $V_\gamma(n+2, n) \leq \mathcal{G}_\gamma(\mathcal{C}_\gamma)$, both cases yield the matching limsup bound. Hence $V_\gamma(n+2, n) = 8 + o(1)$, proving (83). Corollary 5, applied with $K = 8$, then gives (84) and (86). Substitution of (67), specialized to $k = 2$, gives (85). ■

Theorem 5 gives the strongest proved separation between fixed and SNR-dependent codebook design in this paper. The optimal leading coefficient is $K_{n+2,n}^* = 8$, independently of n , whereas the best fixed-codebook leading coefficient is $K_{n+2,n}^{\text{fix}} = 4n$. For $n \geq 3$, the moving family genuinely attains the optimal leading coefficient, and its asymptotic ML error is smaller by the

factor $n/2$ than the best asymptotic ML error of any fixed packing-optimal codebook; no fixed packing-optimal codebook can attain $K_{n+2,n}^*$. For $n = 2$, the two endpoints coincide and the fixed square is already optimal.

D. Optimal Versus Best Fixed-Codebook Leading Coefficients

Combining Proposition 5 with the fixed benchmark in (68) gives, for every fixed $n \geq 3$ and $2 \leq k < n$,

$$K_{n+k,n}^* \leq 4k(k-1) < 4n(k-1) = K_{n+k,n}^{\text{fix}}. \quad (87)$$

In particular,

$$\frac{K_{n+k,n}^*}{K_{n+k,n}^{\text{fix}}} \leq \frac{k}{n} < 1. \quad (88)$$

Let $\mathcal{C}_0^{\text{fix}}$ be any fixed packing-optimal codebook attaining $K_{n+k,n}^{\text{fix}}$, and let $\mathcal{C}_\gamma$ be the family in (72). The fixed-codebook expansion, (77), and (43) yield

$$\begin{aligned} \frac{P_e(\mathcal{C}_\gamma; \gamma)}{P_e(\mathcal{C}_0^{\text{fix}}; \gamma)} &\longrightarrow \frac{k}{n}, \\ \frac{P_e(\mathcal{C}_0^{\text{fix}}; \gamma)}{P_e^*(n+k, n; \gamma)} &\longrightarrow \frac{K_{n+k,n}^{\text{fix}}}{K_{n+k,n}^*} \geq \frac{n}{k}. \end{aligned} \quad (89)$$

Thus, throughout the nonterminal orthoplex-bound range, the explicit moving family has a leading coefficient equal to the fraction k/n of the best fixed-codebook leading coefficient. The exact SNR-wise optimum is no larger than the error of this family.

At $k = 2$, Theorem 5 shows that the bound is sharp:

$$\frac{K_{n+2,n}^*}{K_{n+2,n}^{\text{fix}}} = \frac{2}{n}, \quad \frac{P_e(\mathcal{C}_0^{\text{fix}}; \gamma)}{P_e^*(n+2, n; \gamma)} \longrightarrow \frac{n}{2}. \quad (90)$$

For $3 \leq k \leq n-1$, only the one-sided upper bound on $K_{n+k,n}^*$ has been proved. At the terminal endpoint, Theorem 4 gives

$$\frac{K_{2n,n}^*}{K_{2n,n}^{\text{fix}}} = 1, \quad (91)$$

so no leading-coefficient gain from an SNR-dependent codebook occurs. In particular, when $n = 2$, the left and terminal endpoints coincide and the fixed square is already leading-order optimal.

For $3 \leq k \leq n-1$, we have not proved a global lower bound matching (76). We therefore cannot exclude a different jointly feasible multiscale deformation with a still smaller value of the sum in (54). Nevertheless, the two exact endpoint results and the structure of the explicit construction suggest that its achieved value is sharp, motivating the following conjecture.

Conjecture 1 (Sharpness in the intermediate orthoplex-bound range). *For every fixed $n \geq 4$ and $3 \leq k \leq n - 1$,*

$$K_{n+k,n}^* = 4k(k - 1). \quad (92)$$

The conjectured range is empty for $n \leq 3$. If the conjecture holds, the moving family in (72) attains the optimal leading coefficient throughout the intermediate range.

Table I separates the proved exact values, the proved upper bound, and the conjectured equality.

TABLE I
STATUS OF THE LEADING COEFFICIENTS IN THE SIMPLEX AND ORTHOPLEX-BOUND APPLICATIONS.

Range	Optimal leading coefficient	Best fixed-codebook leading coefficient
$2 \leq M \leq n + 1$	$M(M - 1)$ (exact)	$M(M - 1)$
$k = 2$	8 (exact)	$4n$
$3 \leq k \leq n - 1$	$\leq 4k(k - 1)$; equality conjectured	$4n(k - 1)$
$k = n$	$4n(n - 1)$ (exact)	$4n(n - 1)$

In the intermediate row, $4k(k - 1)$ is the leading coefficient of the constructed family and hence an upper bound on $K_{n+k,n}^*$; equality is conjectured.

VIII. CONCLUSION

Optimal spherical packing fixes the best high-SNR error exponent, but it need not fix the optimal leading coefficient when the codebook is reoptimized at each SNR. The nonterminal orthoplex-bound construction exposes the reason. It slightly reduces the separations of selected pairs by an amount negligible at the leading-coefficient scale, while using the resulting geometric freedom to improve other pairs that become closest in the limiting packing on a larger, though still vanishing, scale. The selected pairs make unit contributions to the constructed family's leading coefficient, whereas the improved pairs make vanishing contributions to that coefficient. The codebooks therefore approach an optimal packing and retain its exponent, yet outperform every packing-optimal codebook held fixed. The improvement results from the two SNR-dependent convergence rates, not from the limiting packing alone.

For every fixed finite (M, n) with $M \geq 2$ and $\rho_{M,n}^* < 1$, we proved

$$P_e^*(M, n; \gamma) = B_\gamma^*(K_{M,n}^* + o(1)).$$

The best packing-optimal benchmark satisfies

$$P_e^{\text{fix}}(M, n; \gamma) = B_\gamma^*(K_{M,n}^{\text{fix}} + o(1)).$$

Moreover, $K_{M,n}^* \leq K_{M,n}^{\text{fix}}$. Thus, whenever $K_{M,n}^* < K_{M,n}^{\text{fix}}$, the exact SNR-wise minimum error is strictly smaller than the best packing-optimal benchmark for all sufficiently large γ , and their ratio tends to $K_{M,n}^*/K_{M,n}^{\text{fix}}$. We showed that the minimum of the Gaussian soft-packing energy converges to $K_{M,n}^*$. We also showed that only pairs that are closest in a limiting packing can contribute to $K_{M,n}^*$ and that their contributions are determined by the pairs' scaled correlation offsets.

The simplex and orthoplex-bound ranges display both possible outcomes. For $2 \leq M \leq n+1$, the regular simplex gives $K_{M,n}^* = K_{M,n}^{\text{fix}} = M(M-1)$, so SNR-dependent motion provides no leading-order gain. For $M = n+k$, $2 \leq k \leq n$, the best fixed-codebook leading coefficient is $4n(k-1)$. When $2 \leq k < n$, the exact ML error of the explicit moving family has leading coefficient $4k(k-1)$, exactly the fraction k/n of the best fixed-codebook leading coefficient; hence

$$\frac{K_{n+k,n}^*}{K_{n+k,n}^{\text{fix}}} \leq \frac{k}{n} < 1.$$

A matching converse proves $K_{n+2,n}^* = 8$ at $k = 2$. At the terminal endpoint $k = n$, the fixed cross-polytope attains the optimal leading coefficient and $K_{2n,n}^* = K_{2n,n}^{\text{fix}} = 4n(n-1)$. For $3 \leq k \leq n-1$, equality in the constructive upper bound remains conjectural.

APPENDIX A

UNIFORM TRANSFER, OPTIMAL LEADING-COEFFICIENT CHARACTERIZATION, AND GAUSSIAN SOFT-PACKING ENERGY ATTAINMENT PROOFS

This appendix proves the uniform-transfer theorem, the $O((n\gamma)^{-1})$ comparison between the normalized optimized exact ML error and $V_\gamma(M, n)$, the existence and characterization of the optimal leading coefficient $K_{M,n}^*$, and the Gaussian soft-packing energy near-minimizer attainment criterion from Section III. The pointwise fixed-codebook expansion is the standard result cited in Subsection II-C and is not rederived here. Instead, the first proof establishes the additional uniformity needed when the codebook changes with SNR. It invokes standard Gaussian-tail and Bonferroni estimates, while deriving only the codebook-uniform estimates specific to the present problem. Set

$$\delta^* = 1 - \rho_{M,n}^* > 0,$$

$$\begin{aligned}\Delta_e(\mathcal{C}) &= \rho_e(\mathcal{C}) - \rho_{M,n}^*, \\ d_e(\mathcal{C}) &= 1 - \rho_e(\mathcal{C}) = \delta^* - \Delta_e(\mathcal{C}).\end{aligned}$$

For $e = (m, \ell)$, abbreviate $d_{m\ell} := d_{(m,\ell)}(\mathcal{C})$, and write

$$w_e(\mathcal{C}, \gamma) = \exp\{n\gamma\Delta_e(\mathcal{C})/4\},$$

so that $\mathcal{G}_\gamma(\mathcal{C}) = \sum_{e \in \mathcal{E}_M} w_e(\mathcal{C}, \gamma)$. Put $N_M = M(M-1)$. Hidden constants in $O_L(\cdot)$ may depend on the fixed pair (M, n) and on L , but not on $\mathcal{C}$ or γ .

A. Uniform Transfer on Bounded-Energy Sublevels

Proof of Theorem 1: Fix $L \geq 1$ and suppose $\mathcal{G}_\gamma(\mathcal{C}) \leq L$. The bounded-energy packing consequence (28) gives

$$0 \leq \rho_{\max}(\mathcal{C}) - \rho_{M,n}^* \leq \eta_\gamma, \quad \eta_\gamma = \frac{4 \log L}{n\gamma}.$$

Hence, for all sufficiently large γ ,

$$\|\mathbf{c}_m - \mathbf{c}_\ell\|^2 = 2d_{m\ell} \geq 2(\delta^* - \eta_\gamma) \geq \delta^*,$$

so the codewords are distinct. If $M \geq 3$, consider the compact set of triples of unit vectors with all mutual distances at least $\sqrt{\delta^*}$. On this set, the inner product of the two unit difference directions from one point is strictly below one: equality would place the other two points on the same ray, but that ray meets the unit sphere in only one further point. Let r denote the maximum over this compact set; the preceding argument shows that $r < 1$. Put $\bar{\varrho} = \max\{0, r\}$. Then, uniformly over this bounded-energy sublevel,

$$\mathbf{u}_{m\ell}^\top \mathbf{u}_{mk} \leq \bar{\varrho}, \quad \mathbf{u}_{m\ell} = \frac{\mathbf{c}_\ell - \mathbf{c}_m}{\|\mathbf{c}_\ell - \mathbf{c}_m\|},$$

for every transmitted index m and distinct competitors $\ell, k \neq m$. For $M = 2$, this assertion is vacuous.

Define the full ordered-pair tail sum

$$\mathcal{Q}_\gamma(\mathcal{C}) = \frac{1}{M} \sum_{e \in \mathcal{E}_M} Q\left(\sqrt{\frac{n\gamma d_e(\mathcal{C})}{2}}\right).$$

The preceding correlation bound gives $d_e(\mathcal{C}) \in [\delta^*/2, 2]$ for all sufficiently large γ . The classical two-sided Mills inequalities [17] imply, uniformly over all ordered pairs,

$$\begin{aligned} & \frac{M^{-1}Q(\sqrt{n\gamma d_e(\mathcal{C})/2})}{B_\gamma^*} \\ &= w_e(\mathcal{C}, \gamma) \left(\frac{\delta^*}{d_e(\mathcal{C})} \right)^{1/2} \\ & \quad \times [1 + O((n\gamma)^{-1})]. \end{aligned}$$

On the stated interval, $|(\delta^*/d_e)^{1/2} - 1| \leq C|\Delta_e|$. For nonnegative offsets,

$$\sum_{\Delta_e \geq 0} w_e |\Delta_e| \leq \eta_\gamma \mathcal{G}_\gamma(\mathcal{C}) = O_L((n\gamma)^{-1}),$$

whereas, for $\Delta_e = -x < 0$, $x \exp\{-n\gamma x/4\} \leq 4/(en\gamma)$. Together with $\sum_e w_e \leq L$, these estimates give

$$\frac{\mathcal{Q}_\gamma(\mathcal{C})}{B_\gamma^*} = \mathcal{G}_\gamma(\mathcal{C}) + O_L((n\gamma)^{-1}). \quad (93)$$

It remains to replace the sum of pairwise tails by the exact union. The union bound and the second Bonferroni inequality [18, Ch. 1] give

$$\begin{aligned} 0 &\leq \mathcal{Q}_\gamma(\mathcal{C}) - P_e(\mathcal{C}; \gamma) \\ &\leq \frac{1}{M} \sum_{m=1}^M \sum_{\substack{\ell < k \\ \ell, k \neq m}} \mathbb{P}(\mathcal{A}_{m\ell} \cap \mathcal{A}_{mk}). \end{aligned}$$

Each event has the form $\mathcal{A}_{m\ell} = \{\mathbf{Z}^\top \mathbf{u}_{m\ell} \geq x_{m\ell}\}$, where

$$x_{m\ell} = \sqrt{\frac{n\gamma d_{m\ell}}{2}} \geq x_{\min} := \sqrt{\frac{n\gamma(\delta^* - \eta_\gamma)}{2}}.$$

If the two normals are antipodal, the corresponding events are disjoint. Otherwise their intersection implies $\mathbf{Z}^\top(\mathbf{u}_{m\ell} + \mathbf{u}_{mk}) \geq 2x_{\min}$, and the standard scalar Gaussian tail bound [1] gives

$$\mathbb{P}(\mathcal{A}_{m\ell} \cap \mathcal{A}_{mk}) \leq \exp \left\{ -\frac{n\gamma(\delta^* - \eta_\gamma)}{2(1 + \bar{\varrho})} \right\}.$$

Because $\bar{\varrho} < 1$, this exponent exceeds the exponent $n\gamma\delta^*/4$ in B_γ^* by a fixed positive multiple of $n\gamma$ for all sufficiently large γ . The number of intersections is fixed, so, for some $\chi_L > 0$,

$$\frac{\mathcal{Q}_\gamma(\mathcal{C}) - P_e(\mathcal{C}; \gamma)}{B_\gamma^*} = O_L(\exp\{-\chi_L n\gamma\}).$$

Combining this estimate with (93) proves the asymptotic form of (32). Enlarging its constant gives the asserted bound for every $\gamma \geq \gamma_L$. ■

B. Packing-Scale Error Implies Bounded Gaussian Soft-Packing Energy

Lemma 2 (Error-to-energy localization). *If an SNR-indexed family $\{\mathcal{C}_\gamma\} \subset \mathcal{M}$ satisfies*

$$\frac{P_e(\mathcal{C}_\gamma; \gamma)}{B_\gamma^*} = O(1),$$

then its codewords are pairwise distinct for all sufficiently large γ , and

$$\rho_{\max}(\mathcal{C}_\gamma) - \rho_{M,n}^* = O((n\gamma)^{-1}), \quad \mathcal{G}_\gamma(\mathcal{C}_\gamma) = O(1). \quad (94)$$

Proof: If two codewords coincide, then for every received vector a decoder can correctly distinguish at most one of their two equiprobable message labels; equivalently, the two corresponding conditional probabilities of correct decoding sum to at most one, independently of the tie-breaking rule. Those two messages therefore contribute at least $1/M$ to the average error. Since $B_\gamma^* \rightarrow 0$, the assumed packing-scale error makes coincident codewords impossible for all sufficiently large γ . Put $\xi_\gamma = \rho_{\max}(\mathcal{C}_\gamma) - \rho_{M,n}^* \geq 0$ and $d_\gamma = \delta^* - \xi_\gamma$. A pair attaining the maximal correlation gives

$$P_e(\mathcal{C}_\gamma; \gamma) \geq \frac{1}{M} Q\left(\sqrt{\frac{n\gamma d_\gamma}{2}}\right).$$

The global lower Gaussian-tail estimate

$$Q(x) \geq c \frac{\exp\{-x^2/2\}}{1+x}, \quad x \geq 0,$$

follows, for example, by combining monotonicity on $0 \leq x \leq 1$ with the classical Mills inequalities on $x \geq 1$ [17]. Since $0 \leq d_\gamma \leq \delta^*$, it gives

$$\begin{aligned} \frac{P_e(\mathcal{C}_\gamma; \gamma)}{B_\gamma^*} &\geq c' \frac{\sqrt{n\gamma}}{1 + \sqrt{n\gamma d_\gamma/2}} \\ &\quad \times \exp\{n\gamma \xi_\gamma/4\} \\ &\geq c'' \exp\{n\gamma \xi_\gamma/4\}. \end{aligned}$$

Bounded normalized error therefore implies $n\gamma \xi_\gamma = O(1)$. Since every offset $\Delta_e(\mathcal{C}_\gamma) \leq \xi_\gamma$,

$$\mathcal{G}_\gamma(\mathcal{C}_\gamma) \leq N_M \exp\{n\gamma \xi_\gamma/4\} = O(1),$$

which proves (94). ■

C. Equivalence of the Optimized Values

Proof of Corollary 2: Let $\mathcal{C}_\gamma^G$ minimize $\mathcal{G}_\gamma$. Evaluation at any fixed packing-optimal codebook gives $V_\gamma(M, n) \leq N_M$, so Theorem 1 yields

$$\frac{P_e^*(M, n; \gamma)}{B_\gamma^*} \leq \frac{P_e(\mathcal{C}_\gamma^G; \gamma)}{B_\gamma^*} = V_\gamma(M, n) + O((n\gamma)^{-1}).$$

For the reverse inequality, choose an exact ML minimizer $\mathcal{C}_\gamma^*$ as in Subsection II-D. Comparison with any fixed $\mathcal{C}_0 \in \mathcal{P}^*$, followed by (15), gives

$$\frac{P_e(\mathcal{C}_\gamma^*; \gamma)}{B_\gamma^*} \leq \frac{P_e(\mathcal{C}_0; \gamma)}{B_\gamma^*} = O(1).$$

Lemma 2 shows that $\mathcal{G}_\gamma(\mathcal{C}_\gamma^*)$ is bounded. Uniform transfer can therefore be applied with one fixed sublevel, giving

$$\begin{aligned} \frac{P_e^*(M, n; \gamma)}{B_\gamma^*} &= \mathcal{G}_\gamma(\mathcal{C}_\gamma^*) + O((n\gamma)^{-1}) \\ &\geq V_\gamma(M, n) - O((n\gamma)^{-1}). \end{aligned}$$

The two bounds prove (34). ■

D. Existence and Characterization of the Optimal Leading Coefficient

Proof of Theorem 2: The compact codebook space $\mathcal{M}$ is semialgebraic, and $(\gamma, \mathcal{C}) \mapsto \mathcal{G}_\gamma(\mathcal{C})$ is definable, with real parameters, in the real exponential field $\mathbb{R}_{\text{exp}}$. Closure under quantification over $\mathcal{M}$ therefore makes the value function $V_\gamma(M, n)$ definable. O-minimality of $\mathbb{R}_{\text{exp}}$ [19] and the one-variable monotonicity theorem [20, Ch. 3, Sec. 1] imply that this function is monotone or constant on some interval (γ_0, ∞) .

For every codebook, $\rho_{\max}(\mathcal{C}) \geq \rho_{M,n}^*$. If the unordered pair $\{m, \ell\}$ attains $\rho_{\max}(\mathcal{C})$, then both ordered terms (m, ℓ) and (ℓ, m) in $\mathcal{G}_\gamma(\mathcal{C})$ are at least one. Thus $V_\gamma(M, n) \geq 2$. Conversely, evaluating the energy at a packing-optimal codebook gives $V_\gamma(M, n) \leq N_M$, because every packing gap is nonpositive. Eventual monotonicity and these uniform bounds prove existence and finiteness of $\lim_{\gamma \rightarrow \infty} V_\gamma(M, n)$, together with (36).

For a fixed $\mathcal{C}_0 \in \mathcal{P}^*$, equation (29) and $V_\gamma \leq \mathcal{G}_\gamma(\mathcal{C}_0)$ give (38). Finally, Corollary 2 gives

$$\frac{P_e^*(M, n; \gamma)}{B_\gamma^*} = V_\gamma(M, n) + O((n\gamma)^{-1}) = K_{M,n}^* + o(1),$$

which identifies the second limit in (35) and proves (37). ■

E. Attainment by Gaussian Soft-Packing Energy Near-Minimization

Proof of Corollary 3: Equations (39) and (35) give $\mathcal{G}_\gamma(\widehat{\mathcal{C}}_\gamma) \rightarrow K_{M,n}^*$; the bound $V_\gamma(M, n) \leq N_M$ shows that this family has uniformly bounded Gaussian soft-packing energy. Uniform transfer then gives $P_e(\widehat{\mathcal{C}}_\gamma; \gamma)/B_\gamma^* \rightarrow K_{M,n}^*$. Dividing by the optimized normalized-error limit in Theorem 2 is legitimate because $K_{M,n}^* \geq 2$, and proves (41). ■

APPENDIX B

PACKING-TAIL LOCALIZATION AND CLOSEST-PAIR CONTRIBUTION CHARACTERIZATION

This appendix proves Lemma 1, Theorem 3, and Proposition 2 from Section IV. Appendix A already supplies error-to-energy localization, uniform transfer, and convergence of the globally minimized Gaussian soft-packing energy. The first two proofs use compactness and separation of finitely many pair correlations; the final proof combines the limiting closest-pair reduction with the global soft-packing energy limit.

At a coarser level, the known Riesz-energy fact that cluster points of fixed-cardinality minimizers are best packings is analogous to the first localization step [15, Prop. 2]. It does not provide the $1/(n\gamma)$ localization rate, the scaled closest-pair contribution formula, or the exact ML conclusions proved here.

Proof of Lemma 1: Condition (44) and Lemma 2 give (45) and $\mathcal{G}_\gamma(\mathcal{C}_\gamma) = O(1)$, and hence $\rho_{\max}(\mathcal{C}_\gamma) \rightarrow \rho_{M,n}^*$. If the distance to $\mathcal{P}^*$ did not tend to zero, compactness would give a convergent subsequence with fixed positive separation from $\mathcal{P}^*$, whereas continuity of $\rho_{\max}$ would place its limit in $\mathcal{P}^*$. This proves (46).

Under the additional hypotheses, membership $\mathcal{C}_{\gamma_j} \in \mathcal{P}^*$ along an unbounded subsequence would give

$$\mathcal{G}_{\gamma_j}(\mathcal{C}_{\gamma_j}) \geq |\mathcal{I}_{\max}(\mathcal{C}_{\gamma_j})| \geq K_{M,n}^{\text{fix}}.$$

The first inequality follows directly from (27), because every ordered closest-pair index contributes one to $\mathcal{G}_{\gamma_j}(\mathcal{C}_{\gamma_j})$; the second is the fixed closest-pair-count characterization (16). Uniform transfer would then give

$$\frac{P_e(\mathcal{C}_{\gamma_j}; \gamma_j)}{B_{\gamma_j}^*} \geq K_{M,n}^{\text{fix}} - O((n\gamma_j)^{-1}),$$

while attainment of the optimal leading coefficient makes this ratio converge to $K_{M,n}^* < K_{M,n}^{\text{fix}}$, a contradiction. This proves (47). ■

Proof of Theorem 3: Finiteness and $\mathcal{C}_\gamma \rightarrow \mathcal{C}_\infty$ give $a > 0$ such that, for all sufficiently large γ , every $e \notin \mathcal{I}_{\max}(\mathcal{C}_\infty)$ satisfies $\rho_e(\mathcal{C}_\gamma) - \rho_{M,n}^* \leq -a$ (vacuously if there is no such index). Since $\rho_{\max}(\mathcal{C}_\gamma) \geq \rho_{M,n}^*$, the largest correlation must therefore be attained within $\mathcal{I}_{\max}(\mathcal{C}_\infty)$, and

$$\max_{e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)} \Delta_{e,\gamma} = \rho_{\max}(\mathcal{C}_\gamma) - \rho_{M,n}^* \geq 0.$$

Equation (28) gives the stated upper bound. Separating the indices in $\mathcal{I}_{\max}(\mathcal{C}_\infty)$ from the remaining pairs gives

$$\begin{aligned} \mathcal{G}_\gamma(\mathcal{C}_\gamma) &= \sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_\infty)} \exp\{n\gamma\Delta_{e,\gamma}/4\} \\ &\quad + O(\exp\{-an\gamma/4\}). \end{aligned}$$

The remainder is identically zero if every ordered pair belongs to $\mathcal{I}_{\max}(\mathcal{C}_\infty)$. Thus (48) holds for some $c > 0$. Uniform transfer and absorption of the exponential remainder into $O((n\gamma)^{-1})$ then prove (49). $\blacksquare$

A. Global Variational Characterization by Scaled Closest-Pair Offsets

Proof of Proposition 2: Fix $\mathcal{C}_0 \in \mathcal{P}^*$, $\lambda \in \mathcal{L}(\mathcal{C}_0)$, and a witnessing sequence from (53). Theorem 3, specifically (48), gives

$$\mathcal{G}_{\gamma_j}(\mathcal{C}_j) \longrightarrow \sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_0)} \exp\{\lambda_e/4\}.$$

Since $V_{\gamma_j}(M, n) \leq \mathcal{G}_{\gamma_j}(\mathcal{C}_j)$ and Theorem 2 gives $V_{\gamma_j}(M, n) \rightarrow K_{M,n}^*$, the sum in (54) is at least $K_{M,n}^*$ for every admissible pair.

For the reverse inequality, start with $\tilde{\gamma}_j = j$ and select an exact soft-packing energy minimizer $\tilde{\mathcal{C}}_j^G$ at each $\tilde{\gamma}_j$. Its energy is

$$\mathcal{G}_{\tilde{\gamma}_j}(\tilde{\mathcal{C}}_j^G) = V_{\tilde{\gamma}_j}(M, n) \leq M(M-1),$$

so compactness of $\mathcal{M}$ gives indices $j_r \rightarrow \infty$ for which $\tilde{\mathcal{C}}_{j_r}^G \rightarrow \mathcal{C}_0$. Relabel this subsequence by $\gamma_r = \tilde{\gamma}_{j_r} = j_r$ and $\mathcal{C}_r^G = \tilde{\mathcal{C}}_{j_r}^G$. Equation (28) and continuity of $\rho_{\max}$ imply $\mathcal{C}_0 \in \mathcal{P}^*$. Corollary 4 then gives a further subsequence and $\lambda \in \mathcal{L}(\mathcal{C}_0)$ such that

$$\sum_{e \in \mathcal{I}_{\max}(\mathcal{C}_0)} \exp\{\lambda_e/4\} = \lim_{r \rightarrow \infty} V_{\gamma_r}(M, n) = K_{M,n}^*.$$

This proves both equality and attainment in (54). Finally, for any sequence witnessing a minimizing pair, $\mathcal{G}_{\gamma_j}(\mathcal{C}_j) \rightarrow K_{M,n}^*$, and the Gaussian soft-packing energy is uniformly bounded along that

sequence. Theorem 1 then gives the limit of the normalized ML error. Dividing it by the corresponding limit of $P_e^*(M, n; \gamma)/B_\gamma^*$ from Theorem 2, which is positive because $K_{M,n}^* \geq 2$, gives the relative-error limit. ■

APPENDIX C

PROOFS FOR THE SIMPLEX AND ORTHOPLEX-BOUND APPLICATIONS

This appendix supplies the derivations specific to the simplex and orthoplex-bound applications. The simplex and terminal-endpoint arguments use the universal-optimality result in [11, Thm. 1.2 and Table 1], while the orthoplex-bound fixed-codebook argument uses the equality-case classification in [12, Thm. 3]; an equivalent characterization is given in [13, Thm. 2.2]. Only the normalization and closest-pair-count consequences needed here are derived. The multiscale moving-family and signed-balance proofs are paper-specific and are retained in full.

A. Simplex Gaussian Soft-Packing Energy Minimum

Proof of Proposition 3: Put $t = n\gamma/4$ and $f_t(s) = \exp\{-ts/2\}$. Since

$$\mathcal{G}_\gamma(\mathcal{C}) = \exp\left\{\frac{tM}{M-1}\right\} \sum_{m \neq \ell} f_t(\|\mathbf{c}_m - \mathbf{c}_\ell\|^2),$$

and f_t is strictly completely monotone, the universal-optimality result in [11, Thm. 1.2, Table 1, and the simplex special case] makes the embedded regular simplex the unique minimizer among distinct M -point configurations. For $n \geq 2$, continuity extends the same lower bound to the full labeled tuple space, including coincident codewords.

For completeness, the equality case on this enlarged space follows from the centroid identity and strict Jensen convexity:

$$\sum_{m \neq \ell} \mathbf{c}_m^\top \mathbf{c}_\ell = \left\| \sum_{m=1}^M \mathbf{c}_m \right\|^2 - M \geq -M.$$

Thus, with $N_M = M(M-1)$,

$$\mathcal{G}_\gamma(\mathcal{C}) \geq N_M \exp\left\{t \left(\frac{1}{N_M} \sum_{m \neq \ell} \mathbf{c}_m^\top \mathbf{c}_\ell + \frac{1}{M-1} \right)\right\} \geq N_M.$$

Attaining the value $M(M-1)$ forces zero centroid and equal off-diagonal correlations, hence correlation $-1/(M-1)$ for every ordered pair. Thus the minimizers are exactly the embedded regular simplices, up to orthogonal transformation and relabeling. The case $n = 1, M = 2$ follows directly by comparing the antipodal and coincident labeled pairs. ■

B. Best Fixed-Codebook Leading Coefficient in the Orthoplex-Bound Range

Proof of Proposition 4: Fix $2 \leq k \leq n$, put $M = n + k$, and let $\mathcal{C}_0 = \{\mathbf{c}_1, \dots, \mathbf{c}_M\} \in \mathcal{P}^*$. Since $\rho_{M,n}^* = 0$, all off-diagonal correlations are nonpositive. Form the graph on $[M]$ in which m and ℓ are adjacent when $\mathbf{c}_m^\top \mathbf{c}_\ell < 0$. Vectors belonging to different connected components are orthogonal.

For a connected component of size s , let r be the rank of its Gram matrix $\mathbf{G}$. The matrix $\mathbf{A} = \mathbf{I}_s - \mathbf{G}$ is nonnegative and irreducible. Since $\mathbf{G} \succeq 0$, the largest eigenvalue of $\mathbf{A}$ is at most one. If the component is linearly dependent, then one is an eigenvalue of $\mathbf{A}$, and the Perron–Frobenius theorem shows that it is simple. Thus every component has nullity either zero or one. This is the connected-component form of the equality-case classification in [12, Thm. 3]; see also [13, Thm. 2.2].

Let ℓ be the number of dependent components. Since the full Gram matrix has rank at most n ,

$$\ell \geq M - n = k.$$

Write the ranks of the dependent components as $r_1, \dots, r_\ell \geq 1$, so that their sizes are $r_1 + 1, \dots, r_\ell + 1$. Let $r_0 \geq 0$ be the total size of the remaining, full-rank components. Then

$$r_0 + \sum_{i=1}^{\ell} (r_i + 1) = M.$$

If N_- denotes the number of ordered pairs having strictly negative correlation, then

$$N_- \leq r_0(r_0 - 1) + \sum_{i=1}^{\ell} r_i(r_i + 1).$$

Indeed, negative-correlation pairs can occur only within connected components, and the right-hand side bounds all ordered pairs within those components.

Merging r_0 into any one of the r_i 's can only increase the right-hand side, because

$$(r_0 + r_i)(r_0 + r_i + 1) - r_0(r_0 - 1) - r_i(r_i + 1) = 2r_0(r_i + 1) \geq 0.$$

For ℓ positive integers with sum $n + k - \ell$, convexity then shows that the sum of $r(r + 1)$ is maximized when one of them equals $n + k - 2\ell + 1$ and the remaining $\ell - 1$ equal one. The resulting upper bound

$$(n + k - 2\ell + 1)(n + k - 2\ell + 2) + 2(\ell - 1)$$

is decreasing over the feasible integers $\ell \geq k$. Therefore,

$$N_- \leq (n - k + 1)(n - k + 2) + 2(k - 1).$$

Every off-diagonal correlation is either negative or zero, and the ordered zero-correlation pairs are precisely the ordered closest-pair indices. Hence

$$\begin{aligned} |\mathcal{I}_{\max}(\mathcal{C}_0)| &= M(M-1) - N_- \\ &\geq (n+k)(n+k-1) - (n-k+1)(n-k+2) - 2(k-1) \\ &= 4n(k-1). \end{aligned}$$

Conversely, take a regular $(n-k+1)$ -simplex and $k-1$ antipodal pairs in mutually orthogonal subspaces. All correlations within each block are strictly negative, all cross-block correlations are zero, and the ordered zero-correlation count is exactly $4n(k-1)$. The codebook is packing-optimal, so minimizing as in (16) proves (68). ■

C. Multiscale Construction in the Nonterminal Orthoplex-Bound Range

Proof of Proposition 5: Fix $n \geq 3$ and $2 \leq k < n$, and put $d = n - k$. The Gram matrix specified by (70) is

$$(1 + q_\gamma)\mathbf{I} - q_\gamma \mathbf{1}\mathbf{1}^\top.$$

Its eigenvalues are $1 + q_\gamma$, with multiplicity $d - 1$, and $1 - (d - 1)q_\gamma$, with multiplicity one. It is therefore positive definite for all sufficiently large γ . Taking its positive-definite square root in a fixed basis of $\mathcal{U}$ produces the continuous realization specified in the body, and $\mathbf{a}_1, \dots, \mathbf{a}_d$ converge to that basis.

For each j ,

$$\mathbf{a}_j^\top \sum_{i=1}^d \mathbf{a}_i = 1 - (d - 1)q_\gamma.$$

The definition of $\mathbf{v}_\gamma$ therefore gives

$$\mathbf{a}_j^\top \mathbf{v}_\gamma = -q_\gamma, \quad \|\mathbf{v}_\gamma\|^2 = \frac{dq_\gamma^2}{1 - (d - 1)q_\gamma} = p_\gamma.$$

Because $\mathbf{v}_\gamma \in \mathcal{U}$ and $\mathbf{f}_i \in \mathcal{U}^\perp$,

$$\|\mathbf{v}_\gamma \pm r_\gamma \mathbf{f}_i\|^2 = p_\gamma + r_\gamma^2 = 1.$$

For sufficiently large γ , $p_\gamma < 1/2$, so the points in (72) are distinct and form a spherical codebook of size $2k + d = n + k$. Since $\mathbf{v}_\gamma \rightarrow \mathbf{0}$, $r_\gamma \rightarrow 1$, and the auxiliary vectors converge to an orthonormal basis of $\mathcal{U}$, this codebook converges to the stated deleted cross-polytope.

Write a shifted point as $\mathbf{x}_{i,s} = \mathbf{v}_\gamma + sr_\gamma \mathbf{f}_i$, where $s \in \{-1, 1\}$. Its pair correlations are

pair class	correlation	ordered count
shifted points on different axes	p_γ	$4k(k-1)$
opposite points on one shifted axis	$2p_\gamma - 1$	$2k$
pairs involving auxiliary points	$-q_\gamma$	$d(d+4k-1)$.

Indeed, $\mathbf{x}_{i,s}^\top \mathbf{x}_{j,t} = p_\gamma$ for $i \neq j$, $\mathbf{x}_{i,1}^\top \mathbf{x}_{i,-1} = 2p_\gamma - 1$, and every off-diagonal pair containing an auxiliary point has correlation $-q_\gamma$. The last count is the sum of $d(d-1)$ ordered auxiliary–auxiliary pairs and $4kd$ ordered auxiliary–shifted pairs. The counts satisfy

$$4k(k-1) + 2k + d(d+4k-1) = (n+k)(n+k-1). \quad (95)$$

For every sufficiently large finite γ , $p_\gamma > 0 > -q_\gamma$ and $2p_\gamma - 1 < 0$. Hence $\rho_{\max}(\mathcal{C}_\gamma) = p_\gamma$, attained by the $4k(k-1)$ ordered cross-axis pairs. Since (66) gives $\rho_{n+k,n}^* = 0$, summing the three pair classes proves (73).

The definition of q_γ gives $n\gamma q_\gamma = \log(n\gamma)$, while

$$n\gamma p_\gamma = \frac{d[\log(n\gamma)]^2}{n\gamma[1 - (d-1)q_\gamma]} = O\left(\frac{[\log(n\gamma)]^2}{n\gamma}\right).$$

This proves (74). In (73), the first term therefore tends to $4k(k-1)$, while the second and third terms vanish. Thus (75) holds.

Finally, $V_\gamma(n+k, n) \leq \mathcal{G}_\gamma(\mathcal{C}_\gamma)$. Taking limits and using Theorem 2 proves (76). The constructed family has uniformly bounded Gaussian soft-packing energy, so Theorem 1 and (75) give (77). ■

D. Terminal Orthoplex Endpoint

Proof of Theorem 4: Put $t = n\gamma/4$. Since $\rho_{2n,n}^* = 0$,

$$\mathcal{G}_\gamma(\mathcal{C}) = \sum_{m \neq \ell} \exp\{t \mathbf{c}_m^\top \mathbf{c}_\ell\}.$$

With $f_t(s) = \exp\{-ts/2\}$, one has $\exp\{t \mathbf{x}^\top \mathbf{y}\} = e^t f_t(\|\mathbf{x} - \mathbf{y}\|^2)$, and f_t is strictly completely monotone. Universal optimality of the full cross-polytope therefore makes it a global soft-packing energy minimizer [11, Thm. 1.2 and Table 1]; continuity extends the lower bound to the labeled tuple space allowing coincidences.

The cross-polytope has $4n(n-1)$ ordered orthogonal pairs and $2n$ ordered antipodal pairs. Their respective contributions to $\mathcal{G}_\gamma(\mathcal{C}_{\text{cross}})$ are 1 and e^{-t} , respectively, proving (78). Theorem 2 and

Proposition 4, specialized to $k = n$, give (79). Corollary 2 together with (67), again specialized to $k = n$, gives (80).

Finally, the fixed cross-polytope is an exact soft-packing energy minimizer at every SNR. Corollary 3, applied with zero objective gap, gives

$$\frac{P_e(\mathcal{C}_{\text{cross}}; \gamma)}{P_e^*(2n, n; \gamma)} \longrightarrow 1,$$

which proves relative-error optimality of this particular fixed codebook. $\blacksquare$

E. Signed-Balance Converse at the Left Endpoint

Proof of Proposition 6: By (66), specialized to $k = 2$, the normalization in $\mathcal{G}_\gamma$ has $\rho_{n+2,n}^* = 0$. Put $t = n\gamma/4$ and $\varrho = \rho_{\max}(\mathcal{C})$. If two codewords coincide, then $\varrho = 1$, and their two ordered terms give

$$\mathcal{G}_\gamma(\mathcal{C}) \geq 2 \exp\{t\} \geq 2 = \frac{8}{(1 + \varrho)^2}.$$

We may therefore assume that the codewords are distinct.

By Radon's theorem [21], after discarding zero weights there are disjoint nonempty index sets $\mathcal{J}_+$ and $\mathcal{J}_-$, positive probability weights $(\alpha_i)_{i \in \mathcal{J}_+}$ and $(\beta_j)_{j \in \mathcal{J}_-}$, and $\boldsymbol{\mu} \in \mathbb{R}^n$ such that

$$\sum_{i \in \mathcal{J}_+} \alpha_i \mathbf{c}_i = \sum_{j \in \mathcal{J}_-} \beta_j \mathbf{c}_j = \boldsymbol{\mu}.$$

For every $i \in \mathcal{J}_+$, the $\mathcal{J}_+$ representation and $\mathbf{c}_i^\top \mathbf{c}_h \geq -1$ give

$$\mathbf{c}_i^\top \boldsymbol{\mu} = \alpha_i + \sum_{\substack{h \in \mathcal{J}_+ \\ h \neq i}} \alpha_h \mathbf{c}_i^\top \mathbf{c}_h \geq \alpha_i - (1 - \alpha_i) = 2\alpha_i - 1.$$

The $\mathcal{J}_-$ representation instead gives

$$\mathbf{c}_i^\top \boldsymbol{\mu} = \sum_{j \in \mathcal{J}_-} \beta_j \mathbf{c}_i^\top \mathbf{c}_j \leq \varrho.$$

Interchanging the two sets gives the analogous bound for each β_j . Hence

$$\|\boldsymbol{\alpha}\|_\infty, \|\boldsymbol{\beta}\|_\infty \leq \frac{1 + \varrho}{2}.$$

Since $\sum_{i,j} \alpha_i \beta_j = 1$, weighted AM–GM applied to

$$\frac{\exp\{t \mathbf{c}_i^\top \mathbf{c}_j\}}{\alpha_i \beta_j} \quad \text{with weight } \alpha_i \beta_j$$

yields

$$S := \sum_{i \in \mathcal{J}_+} \sum_{j \in \mathcal{J}_-} \exp\{t \mathbf{c}_i^\top \mathbf{c}_j\}$$

$$\begin{aligned}
&\geq \frac{\exp\{t\|\boldsymbol{\mu}\|^2\}}{\prod_{i \in \mathcal{J}_+} \alpha_i^{\alpha_i} \prod_{j \in \mathcal{J}_-} \beta_j^{\beta_j}} \\
&\geq \frac{1}{\|\boldsymbol{\alpha}\|_\infty \|\boldsymbol{\beta}\|_\infty} \geq \frac{4}{(1 + \varrho)^2}.
\end{aligned}$$

Here $\exp\{t\|\boldsymbol{\mu}\|^2\} \geq 1$. Moreover, $\alpha_i \leq \|\boldsymbol{\alpha}\|_\infty$ and $\sum_i \alpha_i = 1$ give

$$\prod_i \alpha_i^{\alpha_i} \leq \|\boldsymbol{\alpha}\|_\infty,$$

and the analogous inequality holds for $\boldsymbol{\beta}$. The full energy contains both ordered orientations of every cross pair, so $\mathcal{G}_\gamma(\mathcal{C}) \geq 2S$, proving (81).

For the asymptotic consequence, let $\mathcal{C}_\gamma^\mathcal{G}$ minimize $\mathcal{G}_\gamma$ and write $\varrho_\gamma = \rho_{\max}(\mathcal{C}_\gamma^\mathcal{G})$. Evaluation at any packing-optimal codebook gives $V_\gamma(n+2, n) \leq (n+2)(n+1)$. Hence (28), with $\rho_{n+2, n}^* = 0$, gives $\varrho_\gamma = O((n\gamma)^{-1})$. Applying the finite bound to $\mathcal{C}_\gamma^\mathcal{G}$ proves (82). ■

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